

MDL Selects the Wolfram Rule: $\mathbb{Z}_7$ Dynamics, Algebraic Structure, and Standard Model Encoding of the GTE Polynomial

Nova Spivack

Independent Researcher

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Abstract

Wolfram’s Physics Project lacks a derivation of which hypergraph rewriting rule nature selects from an astronomically large space. We provide the missing principle: the Minimum Description Length (MDL) criterion combined with the Perfect Self-Containment (PSC) orbit constraint uniquely selects $p(L, C, R) = C + R - CR - LCR \pmod{7}$ from 7^{343} candidates (Theorem T96-02, Lean 4, zero `sorry`). A Direct-Interpolation Lift Theorem proves the GTE generation orbit’s total-parity shadow forces exactly one multilinear rule: p itself, with Rule 110 as its binary corollary ($\mathbf{A}_{\text{Lean}}$). A sparsity-floor theorem shows any $\text{GF}(7)$ polynomial of per-variable degree ≤ 6 reproducing Rule 110 has at least four monomials, with equality at p ; multilinearity is forced, not assumed.

The invariant-subset lattice of p over $\text{GF}(7)$ is exactly $\{\mathbf{0}\}$, $\{0, 1\}$, and $\mathbb{Z}_7$: Rule 110 is the unique maximal proper invariant sub-CA ($\mathbf{A}_{\text{Lean}}$). The Standard Model family count $N_{\text{fam}} = 5$, a quadratic non-residue mod 7, is the algebraic obstruction to any intermediate sublattice ($\mathbf{A}_{\text{Lean}}$). The master quadratic $m(x) = x^2 + x - 1$ (discriminant 5) governs dual arithmetic completions: split over $\mathbb{R}$ (golden fixed point, electroweak-scale derivation) and inert at 7 (binary floor protection). The Vacuum Uniqueness Theorem ($\mathbf{A}_{\text{Lean}}$) proves the vacuum is the unique temporally fixed configuration at every ring size; Artin–Mazur zeta functions factorize exactly. Eisenstein arithmetic identifies F_{21} as a norm-7 residue-field quotient; the golden and Eisenstein quadratics are the two subfields of $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$, conductor $15 = N_{\text{gen}}N_{\text{fam}}$. A chirality census shows the vacuum-transparent survivor set over all orderings of the five-family ring is exactly $\{\text{Rule 110, Rule 124}\}$, with ordering reversal exchanging the pair ($\mathbf{A}_{\text{Lean}}$). The period-475 attractor of the 5-cell ring factorizes as $5 \times 5 \times 19$, with 19 enumerative rather than mechanism-forced. A 7-cell kink-injection experiment constitutes the falsifiable prediction: forced kink propagation under p with pass criterion $\xi_{\text{kink}} = 7$ cells, verifiable in silico ($\mathbf{A}/\mathbf{D}$).

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Claim-Strength Taxonomy

Every result in this paper carries one of five certification levels:

- **A_{Lean} (A-Lean)** — Lean 4 machine proof, zero **sorry** in **A_{Lean}** results and zero custom axioms in any **A_{Lean}** proof chain. The strongest certification.
- **A/D (A/D)** — Analytic or fully decidable proof; accessible to human verification.
- **A (A)** — Computationally verified result; output independently reproducible from published scripts.
- **B (B)** — Theoretically motivated conjecture with supporting numerical evidence.
- **D (D)** — Speculative hypothesis; requires further development before quantitative claims can be made.

What this paper claims

- **MDL uniqueness of p (T96-02, A_{Lean})**: PSC+MDL uniquely selects $p(L, C, R) = C + R - CR - LCR \pmod{7}$ from 7^{343} Wolfram rule-space candidates; every link machine-certified in Lean 4, zero **sorry**.
- **Direct-Interpolation Lift and MDL Sparsity Floor (A_{Lean})**: The GTE orbit's total-parity shadow pins p as the unique multilinear GF(7) rule; any degree- ≤ 6 polynomial reproducing Rule 110 has at least four monomials, with equality at p .
- **Chirality census (A_{Lean})**: The vacuum-transparent survivor set over all 120 five-family ring orderings is exactly {Rule 110, Rule 124}; ordering reversal exchanges the pair.
- **Vacuum Uniqueness and dynamical zeta (A_{Lean})**: $\text{Fix}(T_n) = \{\text{vacuum}\}$ for every ring size n ; the Artin–Mazur zeta of p factorizes exactly via the golden Möbius map $\mu(x) = (1 + x)^{-1}$.
- **Golden-Quadratic Duality (A/D; components A_{Lean})**: The quadratic $m(x) = x^2 + x - 1$ governs both the SRRG electroweak fixed point and the QNR binary-floor protection; $N_{\text{fam}} = 5$ is the algebraic obstruction to any intermediate invariant sublattice.
- **Period-475 attractor and kink-injection prediction (A/D)**: The 5-cell ring attractor factors as $5 \times 5 \times 19$ with 19 enumerative and non-mechanism-forced; the MDL-saturated reading pre-registers $\xi_{\text{kink}} = 7$ cells as a falsifiable in-silico pass criterion.
- **Gaussian face lemma and conductor-20 Artin class (A/D; finite A_{Lean})**: All admissible primes satisfy $q \equiv 3 \pmod{4}$; the class is exactly residues $\{3, 7\} \pmod{20}$ in $\mathbb{Q}(i, \sqrt{5})$, the Gaussian twin of the conductor-15 class; certified exhaustively for all 248 admissible primes below 10^4 (**gaussian_face_lemma**, A_{Lean}).
- **Klein quartic genus = N_{gen} (A/D; A_{Lean} cert)**: The unique Hurwitz-maximal Riemann surface for $\text{PSL}(2, 7)$ has genus $3 = N_{\text{gen}}$, with F_{21} as a Borel-parabolic subgroup (**klein_quartic_genus_eq_n_gen**, A_{Lean}).

What this paper does not claim

- **Full Lorentz reconstruction from p alone.** SM particles are BPS kinks of Φ_{MDL} ; Object 0 certifies the sub-Compton vacuum texture, not dynamics at or above the Compton scale.
- **Fine mixing angles from the 1D rule.** CKM and PMNS parameters require the GTE cascade (f_{MDL} , Object 1) and the Algebraic Lifting Theorem; they are not derivable from the polynomial Object 0 alone.
- **The $V-A$ chiral origin.** Weak-isospin handedness comes from the CMCA orbit structure; the $\mathbb{Z}_7$ dynamics of p do not directly reproduce the $\text{SU}(2)_L$ projection.
- **Particle identity beyond orbital quantum numbers.** p certifies the algebraic skeleton; physical particle identification requires the PSC-projection layer (f_{MDL}) and the Algebraic Lifting Theorem of Object 2.
- **A complete Standard Model from p alone.** p is Object 0 in the four-object GTE tower; the full SM spectrum emerges at the Φ_{MDL} level (Object 2) via the Algebraic Lifting Theorem.

The Two Levels of the Framework

Level 1 (certificate): $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$ — the 19-bit polynomial that certifies which structures physics must contain.

Level 2 (field): $\mathcal{L}_{\Phi_{\text{MDL}}} = \frac{1}{2}(\partial_\mu \Phi)^2 - \frac{m^2}{49}(1 - \cos 7\Phi) + \mathcal{L}_{\text{gauge}} + \mathcal{L}_\chi$ — the unique continuum field realising the certificate.

The bridge: $p(x, x, x) = x \Rightarrow m_\varphi = m_\tau$. The Level 2 field law is derived in P42 [14]; full derivation: P48 [8].

1 Introduction

Wolfram’s Physics Project [23, 24] proposes that the fundamental laws of physics are encoded in a specific hyperedge rewriting rule applied iteratively to a spatial hypergraph. This framework connects computational universality, causal invariance, and the emergence of Lorentz symmetry in a compelling unified picture. A critical gap, however, remains open: Wolfram’s programme has no derivation of *which* rule nature selects. The search is empirical — running millions of rules, looking for behavioral richness — across a rule space of size k^{k^3} . For $k = 7$ alone this is $7^{343} \approx 10^{290}$ candidates, far beyond any exhaustive search.

The GTE (Generative Triple Evolution) / UGP programme fills this gap with a theorem. Theorem T96-02 proves that the Minimum Description Length (MDL) principle, combined with the Perfect Self-Containment (PSC) orbit constraint — the requirement that the system’s evolution is fully self-referential: all information needed to specify the dynamics is encoded within the orbit itself, with no external observer or undetermined boundary condition [10, 12] — uniquely selects $p(L, C, R) = C + R - CR - LCR \pmod{7}$ from the entire Wolfram rule space. The selection chain is: PSC forces $k = 7$ (no three-step kink orbit exists for $k \in \{2, 3, 5\}$); MDL then selects p as the unique polynomial-representable rule within the $k = 7$ class. The 19-bit polynomial description achieves a 944-bit savings over a full $k = 7$ lookup table. Every link of this chain is machine-certified in Lean 4 (zero sorry): the cross-family coding step as a Lean coding inequality, and the within- $k=7$ uniqueness by the Direct-Interpolation Lift and sparsity-floor theorems established in this paper (§4.4); §3 states the precise certified content of each link [13].

This paper presents the first systematic study of $p(L, C, R) \pmod{7}$ as a standalone dynamical system. Prior GTE papers [20, 21] studied the CMCA (f_{MDL} , Object 1), whose Rule 110 sublayer

and 10 SM-orbit entries have been used to derive special relativity, gravity, quantum mechanics, and particle masses. Here we study Object 0: the raw polynomial p , run on all 343 inputs over $\text{GF}(7)$. Objects 0 and 1 share only their binary sublayer (Rule 110); they are otherwise algebraically distinct — f_{MDL} is provably not a polynomial over $\text{GF}(7)$ by the Schwartz–Zippel theorem [7, 25], while p is the unique degree- ≤ 3 polynomial satisfying the MDL and PSC criteria.

The paper is organized as follows. Section 2 introduces the four-object GTE derivation tower. Section 3 states T96-02 and the MDL table. Section 4 develops the algebraic structure of p : the binary restriction, invariant-subset and QNR floor theorems, the Direct-Interpolation Lift from the UGP generation orbit and its chirality census (§§4.4–4.5), the golden-quadratic duality (§4.7), the Eisenstein arithmetic of $V(p)$ and the GTE constants (§4.10), the full symmetry group $\text{AGL}(1, 7)$ (§4.11), and the Walsh nonlinearity formula. Section 5 encodes p within Wolfram’s formalism, including the causal graph and a formal botanical correspondence between its fractal binary tree structure and Apiaceae compound umbel L-system models (§5.4). Sections 6–7 present the first spacetime diagrams, behavioral classification, vacuum uniqueness and dynamical zeta structure (§§6.6–6.8), the directed interface spectroscopy of the pair digraph (§6.9), and SM generation orbit of Object 0. Section 8 derives the 3+1D bulk causal graph. Section 9 clarifies what Object 0 is and is not physically. Section 10 lists open questions. Appendices provide Lean certification tables and reproducibility instructions.

Main contributions.

- **T96-02 ($\mathbf{A}_{\text{Lean}}$ chain):** MDL + PSC uniquely selects $p(L, C, R) = C + R - CR - LCR \pmod{7}$ from $7^{343} \approx 10^{290}$ Wolfram rule-space candidates — the selection principle Wolfram’s programme lacks. Every link of the chain is machine-certified in Lean 4: the cross-family coding inequality (`mdl_ca_rule_coding_closed`) and the within- $k=7$ uniqueness (the Direct-Interpolation Lift and sparsity-floor theorems below).
- **Direct-Interpolation Lift ($\mathbf{A}_{\text{Lean}}$):** The total-parity shadow of the canonical GTE cascade triples on the five-family ring, together with vacuum transparency, pins exactly one rule among all $7^8 = 5,764,801$ multilinear $\text{GF}(7)$ candidates: p itself. Rule 110 is thereby *derived* as the binary restriction — a corollary, not an anchor. The orbit input is itself certified cascade output (`gte_orbit_parity_provenance`); the lift is certified by two independent routes, an exhaustive 7^8 census and a structural Möbius-inversion argument (§4.4).
- **MDL sparsity floor ($\mathbf{A}_{\text{Lean}}$):** Any $\text{GF}(7)$ polynomial of per-variable degree ≤ 6 (a class containing a representative of every polynomial function on $\text{GF}(7)^3$) whose binary restriction is Rule 110 flattens to p under exponent flattening and has at least four monomials, with equality exactly at p : multilinearity is MDL-forced, not declared (§4.4).
- **Chirality census ($\mathbf{A}_{\text{Lean}}$):** Over all 120 orderings of the five families on the ring, exactly 20 admit an orbit-consistent elementary CA rule, and the vacuum-transparent survivor union is exactly {Rule 110, Rule 124}, with ordering reversal exchanging the two: the unordered UGP data forces the CMCA chiral pair, and the canonical ordering is the chirality gauge (§4.5).
- **Four-object derivation tower:** The GTE framework produces four distinct objects — p (Object 0, polynomial), f_{MDL} (Object 1, non-polynomial lookup), Φ_{MDL} (Object 2, continuum field), and T (Object T, integer arithmetic cascade). Objects 0 and T are co-selected siblings; T is proved non-polynomial over $\text{GF}(7)$ ($\mathbf{A}_{\text{Lean}}$, `T_b_output_not_determined_by_mod7`, `UgpLean/Polynomial/PolyExplorations.lean`).
- **Multilinear extension (A/D):** p is the unique multilinear lift of Rule 110 to all commutative rings (Theorem 4.14).

- **Cyclotomic point count (A/D):** $|V(p)(\text{GF}(q))| = q^2 - q + 1 = \Phi_6(q)$ for every prime power q ; $V(p)$ is a rational variety with zeta function $(1 - qt)/((1 - t)(1 - q^2t))$ (Theorem 4.15).
- **Binary restriction ($\mathbf{A}_{\text{Lean}}$):** p restricted to binary inputs is exactly Rule 110; thus the Wolfram ether IS the $\{0, 1\}$ -sublayer of a $k=7$ polynomial.
- **Invariant subset uniqueness ($\mathbf{A}_{\text{Lean}}$):** The only non-empty closed sub-alphabets of p over $\text{GF}(7)$ are $\{0\}$, $\{0, 1\}$, and $\mathbb{Z}_7$. Rule 110 is the unique maximal proper invariant sub-CA.
- **Polynomial uniqueness ($\mathbf{A}_{\text{Lean}}$):** p is the unique multilinear polynomial over $\text{GF}(7)$ reproducing Rule 110 on binary inputs — machine-certified over the full 7^8 coefficient class, upgrading the earlier 7^4 form-class verification (§4.4).
- **QNR binary floor ($\mathbf{A}_{\text{Lean}}$):** The uniqueness of $\{0, 1\}$ as the binary floor is explained by $N_{\text{fam}} = 5$ being a quadratic non-residue mod 7 — the SM fermion family count is the algebraic obstruction to any intermediate invariant sublattice.
- **Golden-Quadratic Duality (A/D; components $\mathbf{A}_{\text{Lean}}$):** The QNR obstruction quadratic $m(x) = x^2 + x - 1$ is the diagonal factor of p over every commutative ring $(p(x, x, x) - x = -x m(x), \mathbf{A}_{\text{Lean}})$; its archimedean root $1/\phi$ is the SRRG fixed point of the electroweak-scale derivation, while its 7-adic inertness (no root mod 7^k for any k , $\mathbf{A}_{\text{Lean}}$) protects the binary floor. An all- q dichotomy theorem, an invariant-subset classification deriving the $\text{GF}(5)$ ballistic pathology from ramification, and an honest monodromy negative complete the picture (§4.7).
- **Vacuum Uniqueness Theorem ($\mathbf{A}_{\text{Lean}}$):** $\text{Fix}(T_n) = \{\text{vacuum}\}$ for *every* ring size n , proved by reducing fixed configurations to the golden Möbius map $\mu(x) = (1 + x)^{-1}$, a single 8-cycle on $\mathbb{P}^1(\text{GF}(7))$; with a general- q criterion verified for all primes $q < 200$ and exact Artin–Mazur zeta factorizations (§§6.6–6.7).
- **Eisenstein arithmetic (A/D; components $\mathbf{A}_{\text{Lean}}$):** $F_{21} \cong (\mathbb{Z}[\omega]/(3+\omega))^+ \rtimes \mu_3$ — the internal symmetry group is the norm-7 residue-field quotient of the Eisenstein integers; the motivic identity $[V(p)] = \Phi_6(\mathbb{L})$; a torus action on $V(p)$ whose unique fixed zero is the ether point; the Φ_6 ladder identity web for the GTE constants; and the biquadratic compositum $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$ of conductor $15 = N_{\text{gen}}N_{\text{fam}}$ unifying the golden and Eisenstein rings (§4.10).
- **AGL(1, 7) chiral $\mathbb{Z}_2$ (A/D; $\mathbf{A}_{\text{Lean}}$ core):** The full symmetry group of $V(p)$ is $\text{AGL}(1, 7) = F_{21} \rtimes \mathbb{Z}_2$ of order 42; the reflection element swaps Rule 110 $\leftrightarrow$ Rule 124 over $\text{GF}(7)$ (zero mismatches), inscribing the CMCA chiral pair in the arithmetic of Object 0 (§4.11).
- **Attractor Factorization Theorem (A/D):** The period-475 attractor of the 5-cell ring factors as $5(\text{ring}) \times 5(\text{drift}) \times 19(\text{drift-invariant inert clock})$, with $T^{95} = \sigma^3$ exact, a drift-cancelled return map of order exactly 19, a $\text{GF}(7^3)$ trace carrier, and a linearization no-go (tangent torsion exactly μ_3); the value 19 is certified enumeratively and shown not to be mechanism-forced (§6.8).
- **Directed interface spectroscopy ($\mathbf{A}_{\text{Lean}}$):** The 49-node pair digraph of p carries exact integer interface data: directed wall energies $E_w(1 \rightarrow 0) = 1$, $E_w(0 \rightarrow 1) = 2$, $E_w(5 \rightarrow 0) = 2$, $E_w(0 \rightarrow 5) = E_w(1 \rightarrow 5) = E_w(5 \rightarrow 1) = 4$ with the composite hub identity $E_w(5 \rightarrow 1) = E_w(5 \rightarrow 0) + E_w(0 \rightarrow 1)$, and bump energies (2, 3, 4). The directional asymmetry is the arithmetic trace of the $\text{AGL}(1, 7)$ chiral $\mathbb{Z}_2$, and the half-integer mean $3/2$ of the $0 \leftrightarrow 1$ wall energies is the gap exponent of the associated spin-7 chain (§6.9).
- **Vacuum basin and PSC projector (A):** On the 5-cell ring, p has exactly 52 vacuum-basin states; f_{MDL} 's 10 SM-orbit entries act as a PSC projector routing the GEN orbit through this

basin.

- **Schwartz–Zippel non-polynomial (A):** f_{MDL} (Object 1) is provably not a polynomial of degree ≤ 3 over $\text{GF}(7)$.
- **40% cross-tape fraction (A/D):** For the three-tape DPP system, the cross-tape (gravitational) causal edge fraction is exactly $(N_{\text{TAPES}} - 1)/(N_{\text{TAPES}} + 2) = 40\%$ for $N_{\text{TAPES}} = 3$.
- **Botanical structural equivalence:** The GTE WolframModel causal graph belongs to the same topological class as the Lindenmayer system $F \rightarrow F[+F][-F]$ at $\theta \approx 25^\circ$, $r \approx 0.70$ — the canonical L-system model of the *Daucus carota* compound umbel (Apiaceae) [5, 6]. Agreement on five independent structural metrics (fractal dimension $D = 1.943$, Strahler number 10, Horton ratio $r_B = 2$ ($\mathbf{A}_{\text{Lean}}$: `perfectTree_horton_ratio`), depth, bilateral symmetry) establishes membership in the same structural class (§5.4).

2 The GTE Derivation Tower

The GTE framework produces four distinct mathematical objects from the PSC + MDL derivation chain:

Object 0 — $p(L, C, R) \pmod{7}$, **the algebraic source.** The polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$ is a Wolfram $k=7, r=1$ string substitution rule by definition. Applied to a tape of $\mathbb{Z}_7$ symbols with periodic boundaries, it produces dense chaotic evolution on generic inputs (300 of 343 values nonzero, Class 3 on $\mathbb{Z}_7$) while supporting an exact binary sublayer (Class 4 on $\{0, 1\}^L$ initial conditions). This is the object studied for the first time in this paper.

Object 1 — f_{MDL} , **the Level 1 algebraic certificate (CMCA).** The CMCA lookup table has 18 explicitly specified entries: 8 binary entries that reproduce Rule 110 exactly, plus 10 SM-orbit entries uniquely forced by PSC orbit consistency ($\text{GEN1} \rightarrow \text{GEN2} \rightarrow \text{GEN3} \rightarrow \text{VAC}$, all Lean-certified). Of these 18, exactly 14 output a nonzero value; the four zero-output entries among the 18 are the orbit entry $f_{\text{MDL}}(2, 2, 1) = 0$ (which encodes $\text{GEN2}[3] = 0$ in the $\text{GEN1} \rightarrow \text{GEN2}$ step) and the three binary Rule 110 entries $f_{\text{MDL}}(0, 0, 0) = f_{\text{MDL}}(1, 0, 0) = f_{\text{MDL}}(1, 1, 1) = 0$. The five $\text{GEN3} \rightarrow \text{VAC}$ transition neighborhoods are not listed explicitly; they all evaluate to zero via the MDL-minimal default completion and form part of the remaining 325 of 343 inputs that map to zero. This sparse table (4.1% nonzero) was studied in prior papers [20, 21] to derive SR, GR, QM, and particle masses.

A critical and non-trivial distinction: f_{MDL} and p agree on binary inputs (Rule 110, 8 triples), but they disagree at the 10 SM-orbit neighborhood triples — f_{MDL} ’s outputs at those positions are *specified* values encoding the PSC orbit, not polynomial evaluations. This has been machine-verified: `p_poly_agrees_fmdl_on_binary` proves agreement on $\{0, 1\}^3$; the theorem `p_poly_agrees_fmdl_at_orbit` is refuted by `decide` (counterexample: $(L, C, R) = (1, 1, 5)$, where $f_{\text{MDL}} = 2$ but $p = 3$). Objects 0 and 1 are bridges joined at Rule 110 and divergent everywhere else.

Object 2 — Φ_{MDL} , **the physical substrate.** The continuum limit of f_{MDL} is a $\mathbb{Z}_7$ -symmetric Klein-Gordon scalar field with potential $V(\Phi) = (m^2/49)(1 - \cos 7\Phi)$. Physical particles are BPS topological kink solitons of Φ_{MDL} — stable, energy-minimising field configurations that interpolate between adjacent $\mathbb{Z}_7$ -degenerate vacua [14] — spanning the Compton scale: $\sim 10^{19}$ – 10^{20} lattice cells under the fine-end (Planck-spacing) working hypothesis, exactly $\lambda_C/a = |\mathbb{Z}_7| = 7$ cells under the MDL-saturated matter-sector reading $a = \hbar c/\Lambda_{\text{GTE}}$ [19]. Special relativity, general relativity, quantum mechanics, and the full SM spectrum are derived at this level [9, 11, 14, 21]. Object 2 is not studied directly in this paper.

Object T — T , the arithmetic cascade. The GTE arithmetic map $T : \mathbb{Z}^3 \times \mathbb{N} \rightarrow \mathbb{Z}^3$ governs the integer triple cascade that selects which generation triples exist. At $n = 10$ steps it produces the canonical orbit $(b_1, b_2, b_3) = (73, 42, 275)$ from the seed $(a, b, c) = (1, 73, 823)$ via alternating odd/even arithmetic steps defined by integer division and remainder (Definition 2.5 of [12]). Object T is primitive recursive, machine-certified (`gte_update_map_nat_computable`, `UgpLean/Universality/GTEComputability.lean`, `ALean`).

Object T is provably non-polynomial over $\text{GF}(7)$ (`ALean`, zero sorry). Counterexample: taking $b = 73$, the inputs $(b, c) = (73, 144)$ and $(73, 151)$ satisfy $c \equiv 4 \pmod{7}$ in both cases, but the output b' satisfies $b' \equiv 1 \pmod{7}$ for $c = 144$ and $b' \equiv 3 \pmod{7}$ for $c = 151$. Two inputs with identical $(b \bmod 7, c \bmod 7)$ produce different $b' \bmod 7$ outputs; a polynomial map over $\text{GF}(7)$ cannot exhibit this behavior. Machine-certified (`ALean`, zero sorry): `T_b_output_not_determined_by_mod7`, `T_computable_not_polynomial_GF7` (`UgpLean/Polynomial/PolyExplorations.lean`).

Objects 0 and T are *co-selected siblings*: both are independently derived from MDL + PSC — p within the space of polynomials over $\text{GF}(7)$, T within the space of arithmetic integer cascades — with no algebraic derivability between them. That all four objects are pairwise distinct is machine-certified (`ALean`, zero sorry): `four_object_GTE_pairwise_distinct` (`UgpLean/Polynomial/PolyExplorations.lean`). Object T’s output seeds $(b_1, b_2, b_3) = (73, 42, 275)$ are the generation-level inputs that Object 1 (f_{MDL}) evolves on the 5-cell ring.

What is not a separate object. Rule 110 is the $\{0, 1\}$ -restriction of p (Lean-certified, Theorem 4.1 below); it is Object 0 restricted to binary inputs, not a fifth object. The 19-bit expression $p(L, C, R) = C + R - CR - LCR$ is simply p described compactly; its description length is the MDL-minimal encoding of the $k=7$ rule table.

Algebraic distinction. Object 0 is provably a degree-3 polynomial over $\text{GF}(7)$ by construction. Object 1 is provably *not* a polynomial of degree ≤ 3 over $\text{GF}(7)$: the Schwartz–Zippel theorem [7, 25] bounds the number of zeros of any degree- ≤ 3 polynomial in 3 variables over $\text{GF}(7)$ at $3 \cdot 7^2 = 147$, but f_{MDL} has 329 zeros ($= 343 - 14$, since only 14 of 343 inputs produce nonzero output). Therefore Objects 0 and 1 are categorically distinct — not merely quantitatively different (87.5% vs. 4.1% nonzero) but different in kind: polynomial vs. provably non-polynomial (`A`).

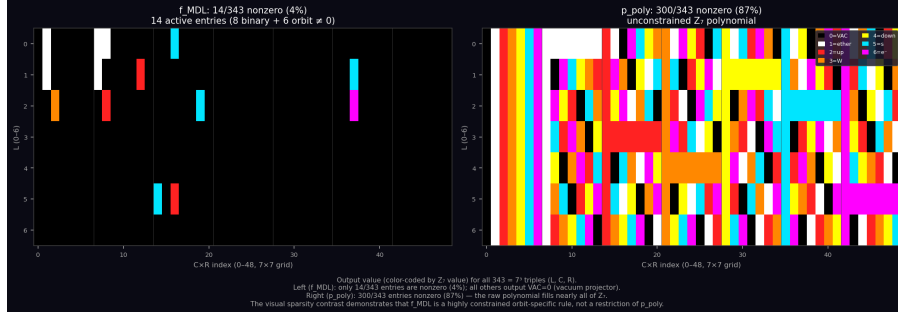


Figure 1: Lookup table comparison: $p(L, C, R) \pmod{7}$ (Object 0, left, 87.5% nonzero) vs. f_{MDL} (Object 1, right, 4.1% nonzero). Each cell represents one of the 343 (L, C, R) triples over $\text{GF}(7)$, colored by output value. The dramatic sparsity difference — combined with the Schwartz–Zippel non-polynomial refutation — establishes that the two objects are categorically distinct. They share exactly the 8 binary triples (Rule 110 sublayer).

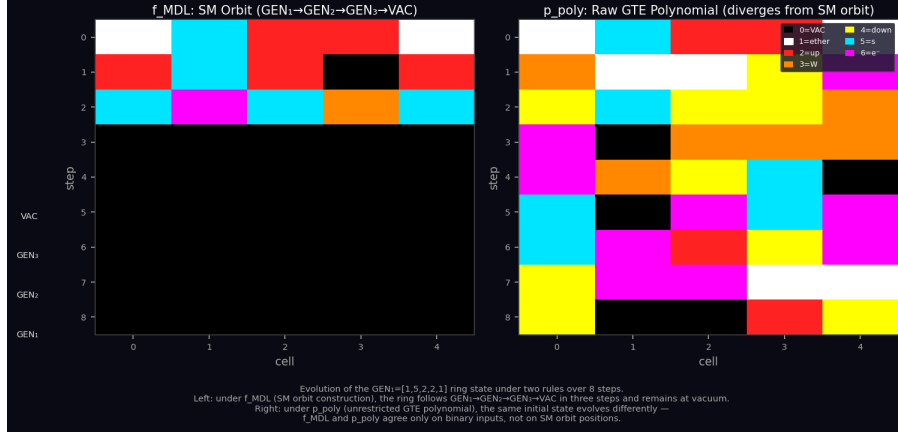


Figure 2: Divergent dynamics of f_{MDL} and p_{poly} starting from $\text{GEN1} = [1, 5, 2, 2, 1]$ on the isolated 5-cell periodic ring. *Left:* Under f_{MDL} , the orbit $\text{GEN1} \rightarrow \text{GEN2} \rightarrow \text{GEN3} \rightarrow \text{VAC}$ completes in exactly three steps (Lean-certified). *Right:* Under p_{poly} , $\text{GEN1} \rightarrow [3, 1, 1, 4, 6]$ at step 1 — diverging immediately from the SM orbit, entering a period-475 attractor in which GEN2 and GEN3 never appear. The SM orbit (Object 1) and the polynomial dynamics (Object 0) are dynamically as well as algebraically distinct.

3 MDL Selects p from Wolfram Rule Space

3.1 Theorem T96-02 ($\mathbf{A}_{\text{Lean}}$)

Theorem 3.1 (T96-02, MDL Selection — $\mathbf{A}_{\text{Lean}}$ chain). *Among all k -symbol, $r=1$ Wolfram string substitution rules, the unique rule satisfying both (i) PSC orbit consistency — admitting a three-step kink orbit $\text{GEN1} \rightarrow \text{GEN2} \rightarrow \text{GEN3} \rightarrow \text{VAC}$ on a 5-cell periodic ring — and (ii) minimum description length (polynomial representability of degree ≤ 3 over $\text{GF}(k)$) is $k = 7$,*

$$p(L, C, R) = C + R - CR - LCR \pmod{7}.$$

The polynomial description requires 19 bits; the explicit $k = 7$ lookup table requires 963 bits. No other k -symbol rule satisfies both criteria.

Certified content, link by link. Theorem 3.1 is a chain, and its certification is stated per link; no single Lean theorem enumerates the 7^{343} class. The cross-family step — the coding inequality by which the $k = 7$ family strictly beats the $k = 5$ alternative, together with the exhaustive $\text{GF}(5)$ kink exclusion — is machine-certified as `mdl_ca_rule_coding_closed` ($\mathbf{A}_{\text{Lean}}$, zero sorry, `UgpLean/Universality/MDLDerivabilityCriterion.lean`). The within- $k=7$ uniqueness is machine-certified by the Direct-Interpolation Lift and sparsity-floor theorems of §4.4 ($\mathbf{A}_{\text{Lean}}$): the PSC orbit enters through its certified total-parity shadow, whose ring evaluations plus vacuum transparency pin the unique multilinear rule (`ugp_orbit_interpolation_lift`), and any $\text{GF}(7)$ polynomial whose binary restriction is Rule 110 carries at least four monomials, with equality exactly at $p(\text{gf7_rule110_sparsity_floor})$ — so the polynomial representability demanded by MDL admits exactly one realization.

The proof proceeds by the following selection chain:

1. *PSC forces $k = 7$.* The canonical PSC orbit $\text{GEN1} \rightarrow \text{GEN2} \rightarrow \text{GEN3} \rightarrow \text{VAC}$ uses six distinct non-vacuum values from $\{1, 2, 3, 5, 6\}$ on a five-cell ring. Concretely: $\text{GEN1} = [1, 5, 2, 2, 1]$,

GEN2 = [2, 5, 2, 0, 2], GEN3 = [5, 6, 5, 3, 5]. The value 6 appears in GEN3 — it must be an element of the field. For $k = 2$: only $\{0, 1\}$ available, orbit cannot close in three steps with three distinct non-trivial states. For $k = 5$: the required value $6 \notin \text{GF}(5)$, and exhaustive machine search (`z5_fmld_no_psc_kink_orbits`, `UgpLean/Universality/MDLDerivabilityCriterion.lean`, zero sorry) confirms no PSC-admissible kink orbit exists. For $k = 7$: all required orbit values $\{0, 1, 2, 3, 5, 6\} \subset \mathbb{Z}_7$; the orbit closes exactly. $k = 7$ is the minimal prime field whose element set contains the full PSC orbit.

2. *MDL selects p within $k = 7$.* Among all $7^{343} \approx 10^{290}$ rules in the $k = 7$ class, the unique polynomial rule (description length 19 bits) consistent with the PSC orbit is $p(L, C, R) = C + R - CR - LCR$. The 19-bit polynomial encoding achieves a 944-bit saving over the explicit 963-bit lookup table. All other $k = 7$ rules either require the full explicit table or are PSC-inconsistent. The uniqueness statement of this step is the certified content of the Direct-Interpolation Lift and sparsity-floor theorems (§4.4).

3.2 The MDL Table

The MDL criterion compares description lengths across symbol counts:

k	Full table (bits)	Polynomial MDL (bits)	PSC status
2	8	8	Fails: no 3-step kink orbit
5	290	smaller	Fails: no PSC kinks (CatAL, exhaustive)
7	963	19	Passes — unique
11	4605	> 19	Fails MDL: longer polynomial description

From $7^{343} \approx 10^{290}$ candidates in the $k=7$ rule space alone, exactly one rule satisfies both MDL and PSC. The full selection chain is: $\text{PSC} \rightarrow k=7 \rightarrow \text{MDL} \rightarrow p$.

3.3 Wolfram (Empirical) vs. GTE (Derived)

Wolfram’s programme runs millions of rules empirically and identifies candidates by behavioral richness [22, 23]. GTE derives the unique candidate from first principles and certifies the derivation in Lean 4. These are complementary: GTE provides the “why this rule”; Wolfram provides the toolset for studying what the rule does. Causal invariance holds trivially for any deterministic CA (including $p \pmod{7}$), consistent with Wolfram’s $\text{CI} \rightarrow \text{SR}$ mechanism [3].

4 Algebraic Structure

4.1 Rule 110 as Binary Restriction ($\mathbf{A}_{\text{Lean}}$)

Theorem 4.1 ($\mathbf{A}_{\text{Lean}}$). *For all $L, C, R \in \{0, 1\}$, $p(L, C, R) \pmod{7} = \text{Rule110}(L, C, R)$. The polynomial p restricted to binary inputs reproduces the Rule 110 truth table on all 8 binary triples. In particular, $p(\{0, 1\}^3) \subseteq \{0, 1\}$, so $\{0, 1\}$ is an invariant subset under p .*

Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry):

- `rule110_z7_poly_rep` — `UgpLean/Universality/PhiMDLUniversality.lean`

- `p_poly_binary_restriction_closed`
`UgpLean/Universality/Z7InvariantSubsets.lean`

The ether background — the stable attractor of Rule 110 initialized on a binary tape [2] — IS the $\{0,1\}$ -sublayer of p . This connection is not a coincidence: the polynomial coefficients $(a, b, c, d) = (1, 1, -1, -1)$ in $p = aC + bR + cCR + dLCR$ are the unique solution over $\text{GF}(7)$ such that this exact identification holds (Theorem 4.3; machine-certified over the full multilinear class in §4.4).

4.2 Invariant Subset Uniqueness ($\mathbf{A}_{\text{Lean}}$)

Theorem 4.2 ($\mathbf{A}_{\text{Lean}}$). *The only non-empty subsets $S \subseteq \mathbb{Z}_7$ such that $p(S \times S \times S) \subseteq S$ are $S = \{0\}$, $S = \{0,1\}$, and $S = \mathbb{Z}_7$. In particular, Rule 110 (the $\{0,1\}$ -restriction of p) is the unique maximal proper invariant sub-CA of p over $\text{GF}(7)$.*

Proof. Exhaustive verification over all $2^7 = 128$ subsets of $\mathbb{Z}_7$ (computational, 202 ms; also certified via `native_decide` in Lean 4). Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry, `native_decide`): `p_poly_invariant_subsets_classification` and `rule110_unique_proper_invariant_subca` in `UgpLean/Universality/Z7InvariantSubsets.lean`. $\square$

Several SM-motivated subset candidates fail: the SM winding values $\{0, 2, 3, 4, 6\}$, the SM-plus-ether set $\{0, 1, 2, 3, 4, 6\}$, and the full GEN orbit values $\{0, 1, 2, 3, 5, 6\}$ are all non-invariant. The witness for $\{0, 2, 3, 4, 6\}$: $p(0, 2, 4) = 5 \notin S$.

Physical interpretation. There is no algebraically closed intermediate CA layer between Rule 110 and full $\mathbb{Z}_7$. Once any $\mathbb{Z}_7$ value beyond $\{0,1\}$ enters the dynamics, the evolution immediately escapes to all of $\mathbb{Z}_7$. This explains the Class 3 chaotic behavior observed in Section 6, the absence of stable $\mathbb{Z}_7$ gliders (Section 6), and why f_{MDL} zeros out all 325 non-orbit triples: there is no intermediate algebraic layer for them to inhabit.

4.3 Polynomial Uniqueness ($\mathbf{A}$)

Theorem 4.3 ($\mathbf{A}$). *Among all polynomials of the form $aC + bR + cCR + dLCR$ over $\text{GF}(7)$ (four free coefficients; $7^4 = 2401$ candidates), the polynomial $C + R - CR - LCR$ (coefficients $a = 1$, $b = 1$, $c = -1$, $d = -1$) is the unique polynomial in this form class that reproduces Rule 110 on all 8 binary inputs $\{0,1\}^3$.*

The proof is by linear algebra over $\text{GF}(7)$: the 8 binary equations $p(L, C, R) = \text{Rule110}(L, C, R)$ for $(L, C, R) \in \{0,1\}^3$ constitute a linear system over $\text{GF}(7)$ whose unique solution (among the $7^4 = 2401$ choices of (a, b, c, d)) is $(1, 1, -1, -1)$, recovering the GTE polynomial. This provides a second, independent axis of uniqueness for p : not only MDL-minimal among all Wolfram $k=7$ rules (T96-02), but also polynomially unique in this degree- ≤ 3 form class. The next subsection upgrades this statement twice over: uniqueness holds over the full eight-coefficient multilinear class (machine-certified), and the input data can be taken from the UGP cascade orbit rather than from the Rule 110 truth table.

4.4 The Direct-Interpolation Lift: p from the UGP Orbit ($\mathbf{A}_{\text{Lean}}$)

The uniqueness statements above take the Rule 110 truth table as input. This subsection inverts the arrow: the polynomial is forced by the arithmetic of the GTE cascade itself, with Rule 110 emerging as a corollary. The input data is the *total-parity shadow* of the canonical generation

triples: the fifteen triples (a, b, c) of the $n = 10$ cascade catalogue (three generations of each of the five SM families) [12, 20] reduce, family by family and generation by generation, to the parities $(a + b + c) \bmod 2$, giving three binary states of the five-family ring. These are exactly the generation-orbit vectors whose two-step evolution characterizes Rule 110 among the 256 elementary CA rules in the computational-universality companion paper [10]; the provenance link — that the cascade arithmetic reproduces those vectors bit for bit — is itself machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry, `gte_orbit_parity_provenance`, with a signed-parity convention covering the top-quark branch).

Theorem 4.4 (Direct-Interpolation Lift — $\mathbf{A}_{\text{Lean}}$). *Consider the $7^8 = 5,764,801$ multilinear polynomials $a_0 + a_L L + a_C C + a_R R + a_{LC} LC + a_{LR} LR + a_{CR} CR + a_{LCR} LCR$ over $\text{GF}(7)$, acting as $r = 1$ CA rules. Impose the ten ring evaluations of the parity-orbit two-step evolution (generation 1 $\rightarrow$ generation 2 $\rightarrow$ generation 3 on the five-cell ring, outputs read as $\text{GF}(7)$ elements) together with vacuum transparency $p(0, 0, 0) = 0$. Exactly one polynomial survives: $p = C + R - CR - LCR$. Without vacuum transparency, exactly seven survive.*

Machine-certified by two independent routes ($\mathbf{A}_{\text{Lean}}$, zero sorry): the exhaustive census over all 7^8 coefficient vectors (`ugp_orbit_interpolation_lift`, `orbit_interpolation_vandermonde_full_rank`, `UgpLean/Universality/TriangleLiftTheorem.lean`), and a structural, enumeration-free route (`orbit_interpolation_lift_structural`, `multilinear_binary_determination`, `orbit_vt_forces_interpolant`, `UgpLean/Universality/TriangleLiftStructural.lean`): the ten orbit evaluations cover seven of the eight binary neighbourhoods, vacuum transparency supplies the eighth, and Möbius inversion of the binary system forces the interpolant.

Two observations keep the hypothesis set honest. First, vacuum transparency is not an extra import: among orbit satisfiers it is equivalent to minimal minterm count — the MDL criterion in the lookup-table metric — machine-certified in the companion paper as the orbit weight dichotomy, `orbit_weight_dichotomy` [10]. Second, the orbit is consumed at the parity level by necessity, not convenience: the identity-mod-7 orbit (the f_{MDL} entries of Object 1) is *inconsistent* on the multilinear class, realizable only at per-variable degree ≥ 2 — a 27-coefficient class costing ≈ 76 bits against 19 bits for the parity shadow. This is the quantitative form of the Object 0 / Object 1 distinction of §2 ($\mathbf{A}$, `scripts/triangle_residual_tests.py`).

Corollary 4.5 (Rule 110 as output — $\mathbf{A}_{\text{Lean}}$). *The binary restriction of the forced interpolant is Rule 110 (`interpolation_lift_binary_corollary`; bundled with the orbit-uniqueness chain as `cup4_parity_uniqueness_from_lift`). In the derivation order of this subsection, Theorem 4.1 is a corollary of the cascade arithmetic rather than an anchor.*

Theorem 4.6 (MDL Sparsity Floor — $\mathbf{A}_{\text{Lean}}$). *Let g be any polynomial over $\text{GF}(7)$ in canonical form (per-variable degree ≤ 6 ; every polynomial function on $\text{GF}(7)^3$ has a representative in this class) whose binary restriction is Rule 110. Then exponent flattening — each positive exponent collapsed to 1, with coefficients summed by support class — sends g to p , and g has at least four nonzero monomials. The floor is attained exactly at $g = p$, whose support classes are $\{C\}$, $\{R\}$, $\{CR\}$, $\{LCR\}$.*

Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry): the canonical-class form as `gf7_rule110_sparsity_floor`, via the flattening homomorphism `gf7_flattening_binary_agreement`, both in `UgpLean/Universality/TriangleLiftStructural.lean`; the multilinear-class form as `rule110_lift_sparsity_floor` in `UgpLean/Universality/TriangleLiftTheorem.lean`.

The floor makes multilinearity MDL-forced rather than declared: any $\text{GF}(7)$ rule whatsoever restricting to Rule 110 pays at least p 's four monomials, and only p pays exactly that. Together,

Theorems 4.4 and 4.6 close the within- $k=7$ link of the T96-02 selection chain (§3) at machine-certified grade.

Remark 4.7 (Scope of the parity reduction — $\mathbf{A}/\mathbf{D}$ on $\mathbf{A}_{\text{Lean}}$ anchors). The lift consumes the *total-parity* shadow of the cascade triples, and this choice is itself forced over the full space in which “reduction of the cascade arithmetic” is meaningful. Exhaustively over all 777 surjective additive reductions $\mathbb{Z}^3 \rightarrow \mathbb{Z}_m$ ($2 \leq m \leq 7$, image embedded in $\text{GF}(7)$) and all $7^5 = 16,807$ mod-2 recodings into $\text{GF}(7)$, the reductions whose induced orbit *coherently* forces a rule — a unique multilinear solution that is closed on the reduction’s image and has a unique homogeneous vacuum, the certified invariant-subalphabet and vacuum-uniqueness conditions of Theorems 4.2 and 6.1 — are exactly total parity up to alphabet units, forcing the unit conjugates $g_t(L, C, R) = t p(t^{-1}L, t^{-1}C, t^{-1}R)$. The exhaustive census counts and survivor identifications are machine-certified (`parity_projection_additive_forcing`, `parity_projection_mod2_recoding_forcing`, `UgpLean/Universality/ParityProjectionForcing.lean`); the assembled scope statement carries grade $\mathbf{A}/\mathbf{D}$ on those anchors and the published batteries ($\mathbf{A}$, `scripts/parity_projection_homomorphism_battery.py`, `scripts/parity_projection_local_function_battery.py`). The canonical multiplicative competitor — product parity $abc \bmod 2$ — forces the *sparser* three-monomial rule $R + LC + 5LCR$, whose binary restriction is Rule 106: naive rule-sparsity is therefore not the reduction-selection principle. The competitor is excluded because its diagonal fixed-point set is $\{0, 4\}$ — a displaced vacuum, the same certified criterion that removes $\text{GF}(5)$ from the modulus chain (Theorem 4.12). Beyond mod-2 granularity, non-additive recodings provably break forcing (the exceptions are classified), and over unrestricted lookup reductions forcing is constructively vacuous: 76 of the 128 vacuum-transparent elementary rules can be painted onto the orbit by an explicit lookup, shadow-closure included ($\mathbf{A}$, `scripts/parity_projection_unrestricted_vacuity.py`). The additive-plus-mod-2 scope is therefore maximal: no forcing theorem over arbitrary recodings can exist, for p or for any other rule.

4.5 The Chirality Census: $\{\text{Rule 110, Rule 124}\}$ from Unordered Orbit Data ($\mathbf{A}_{\text{Lean}}$)

The lift consumes the parity orbit in its canonical family ordering. The census quantifies over the ordering itself.

Theorem 4.8 (Chirality Census — $\mathbf{A}_{\text{Lean}}$). *Over all $5! = 120$ orderings of the five families on the ring: exactly 20 orderings admit an elementary CA rule consistent with the two-step parity orbit; all 20 admit a vacuum-transparent rule; the union of vacuum-transparent survivors over all orderings is exactly $\{\text{Rule 110, Rule 124}\}$; and ordering reversal bijects the ten Rule-110 orderings onto the ten Rule-124 orderings.*

Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry, `native_decide` over 120×256): `orbit_chirality_census`; the link to the certified reflection identity (Rule 124 is Rule 110 reflected) is `orbit_chirality_census_reflection_link` (`UgpLean/Universality/TriangleLiftTheorem.lean`).

The reading: the unordered UGP orbit data forces precisely the chiral outer pair of the CMCA architecture [20, 21], and choosing the canonical ordering is choosing the chirality gauge. This is the census-level face of the variety-level reflection of Theorem 4.25: there, the $\text{AGL}(1, 7)$ reflection swaps Rule 110 with Rule 124 over $\text{GF}(7)$; here, the same $\mathbb{Z}_2$ appears as ordering reversal on the family ring. No claim is made that either orientation is dynamically preferred: the pair is forced; the orientation is conventional. All statements in this and the preceding subsection are finite algebraic-certificate facts about Object 0 and the cascade’s parity data; no continuum-level claim is involved.

4.6 QNR Binary Floor Uniqueness ($\mathbf{A}_{\text{Lean}}$)

Theorem 4.9 ($\mathbf{A}_{\text{Lean}}$). *The binary sublayer $\{0, 1\}$ of p over $\text{GF}(q)$ is the unique non-trivial proper invariant subset if and only if $N_{\text{fam}} = 5$ is a quadratic non-residue mod q , i.e., $q \equiv \pm 2 \pmod{5}$.*

For $\text{GF}(7)$: $7 \equiv 2 \pmod{5}$, so 5 is a QNR mod 7 ($\sqrt{5} \notin \mathbb{Z}_7$). This rules out any intermediate invariant substructure between $\{0, 1\}$ and $\mathbb{Z}_7$ by the following argument. If an element $k \in \mathbb{Z}_7 \setminus \{0, 1\}$ were to anchor an invariant subset beyond $\{0, 1\}$, then in particular $p(k, k, k)$ must lie in that subset. A necessary condition is the constant fixed-point equation $p(k, k, k) = k$: substituting into $p(L, C, R) = C + R - CR - LCR$ gives $k + k - k^2 - k^3 = k$, i.e. $k^3 + k^2 - k = 0$, which factors as $k(k^2 + k - 1) = 0$. The non-trivial factor is $k^2 + k - 1 = 0$, a quadratic whose discriminant is $\Delta = 1^2 - 4(1)(-1) = 5 = N_{\text{fam}}$. Since 5 is a QNR mod 7, this quadratic has no solution in $\mathbb{Z}_7$; hence no $k \notin \{0, 1\}$ can be a constant fixed point, and no intermediate invariant subset exists.

Machine-certified for $q = 7$ ($\mathbf{A}_{\text{Lean}}$, by `decide`, zero sorry):

- `five_is_qnr_mod7, kink_fixed_point_eq_no_solution`
- `no_intermediate_fixed_point, nfam_qnr_explains_binary_floor`

All in `UgpLean/Universality/Z7InvariantSubsets.lean`.

Physical interpretation. $N_{\text{fam}} = 5$ is the Standard Model fermion family count, derived independently from the $\mathbb{Z}_5$ orbit structure of the GTE gauge sector [10, 20]. Theorem 4.9 states that the SM family count IS the algebraic obstruction to any intermediate invariant sub-CA. The reason Rule 110 is the unique binary floor of p is that the physical fermion family number 5 is a quadratic non-residue mod 7. This is a direct connection between the Standard Model generation count and the algebraic structure of $\text{GF}(7)$.

4.7 The Golden-Quadratic Duality

The quadratic $m(x) = x^2 + x - 1$ appearing in the proof of Theorem 4.9 is not specific to $\text{GF}(7)$: it is the non-trivial factor of p 's diagonal self-consistency equation over *every* commutative ring,

$$p(x, x, x) - x = -x(x^2 + x - 1), \quad (1)$$

machine-certified as a universal ring identity ($\mathbf{A}_{\text{Lean}}$, `gte_diagonal_quadratic_factorization`, `UgpLean/Polynomial/GoldenQuadratic.lean`). We call m the *master quadratic* of p . Its discriminant is $\Delta = 5 = N_{\text{fam}}$, and $\mathbb{Z}[x]/(m) \cong \mathbb{Z}[\varphi]$ is the ring of integers of $\mathbb{Q}(\sqrt{5})$ (the golden ring) [4]. This single $\mathbb{Z}$ -quadratic turns out to control two previously separate GTE derivation chains — the electroweak-scale fixed point and the unique binary computational floor — as the behavior of m at two different completions of $\mathbb{Q}$. All statements in this subsection are arithmetic facts about p 's diagonal quadratic (Level 1 algebraic-certificate results); the archimedean fiber is the algebraic *input* to the continuum electroweak-scale derivation, not itself a continuum-field theorem.

Theorem 4.10 (Golden-Quadratic Duality — $\mathbf{A}/\mathbf{D}$; components $\mathbf{A}_{\text{Lean}}$). *Let $m(x) = x^2 + x - 1$, so $\mathbb{Z}[x]/(m) \cong \mathbb{Z}[\varphi]$ with discriminant $5 = N_{\text{fam}}$.*

- (a) Identification. *The SRRG fixed-point equation $g^2 + g - 1 = 0$ of the Higgs-VEV derivation [15, 18] and the constant-fixed-point obstruction of Theorem 4.9 are the images of the single $\mathbb{Z}$ -form m under $\mathbb{Z}[x] \rightarrow \mathbb{R}[x]$ and $\mathbb{Z}[x] \rightarrow \text{GF}(7)[x]$ respectively.*
- (b) Archimedean fiber (split). *Over $\mathbb{R}$, $m = (x - 1/\phi)(x + \phi)$ with ϕ the golden ratio; the positive root $1/\phi$ is the SRRG adjudication fixed point that seeds the electroweak-scale derivation ($\mathbf{A}_{\text{Lean}}$ `gte_poly_srrg_bridge`; see [18]).*

- (c) 7-adic fiber (inert). m is irreducible over $\mathbb{Q}_7$: it has no root in $\text{GF}(7)$, in $\mathbb{Z}/7^k\mathbb{Z}$ for any $k \geq 1$, or in $\mathbb{Q}_7$. The prime 7 is inert in $\mathbb{Z}[\varphi]$ with residue field $\text{GF}(49)$. Machine-certified for all 7-adic depths (`ALean, master_quadratic_no_root_mod_seven_pow`).
- (d) Splitting law. For any prime q , the splitting of m over $\text{GF}(q)$ is governed by $q \bmod 5$ (Theorem 4.11 below); m ramifies exactly at $q = 5$.

Proof sketch. (a) is Eq. (1) read in the two target rings. (b) is the quadratic formula plus the Lean-certified SRRG bridge. (c): no root mod 7 by Theorem 4.9; a root mod 7^{k+1} reduces to a root mod 7^k , so induction excludes every depth; a $\mathbb{Q}_7$ root must have valuation 0 (negative valuation makes $v(m(x)) = 2v(x)$ finite and non-zero, positive valuation makes $m(x) \equiv -1 \pmod{7}$), hence reduces to a root mod 7 — contradiction. (d) is Theorem 4.11. $\square$

The duality bundle — one quadratic, split at the archimedean place where the continuum fixed point must exist, inert at the alphabet prime where any root would create a frozen rival vacuum and break floor uniqueness — is certified as a single Lean statement (`ALean, golden_floor_duality_bundle`). Part (c) strengthens Theorem 4.9: the binary floor is robust not merely mod 7 but at every 7-adic depth.

Theorem 4.11 (All- q Dichotomy — **A/D**). For a prime q :

- (i) $q \notin \{2, 5\}$: m has roots in $\text{GF}(q) \iff \left(\frac{5}{q}\right) = +1 \iff q \equiv \pm 1 \pmod{5}$; when they exist the roots are $2^{-1}(-1 \pm s)$ with $s^2 = 5$, and they are distinct.
- (ii) $q = 5$: m ramifies, $m \equiv (x + 3)^2 \pmod{5}$, with double root $k = 2$.
- (iii) $q = 2$: $m \equiv x^2 + x + 1$ is irreducible.

Proof. For odd q , multiplying by 4 and completing the square gives $4m(x) = (2x + 1)^2 - 5$, and $x \mapsto 2x + 1$ is a bijection of $\text{GF}(q)$; so m has roots iff 5 is a square in $\text{GF}(q)$. Since $(5 - 1)/2 = 2$ is even, quadratic reciprocity [4] gives $\left(\frac{5}{q}\right) = \left(\frac{q}{5}\right)$, and the squares mod 5 are $\{1, 4\}$, yielding the $q \equiv \pm 1 \pmod{5}$ criterion. At $q = 5$ the discriminant vanishes and $m \equiv (x + 3)^2$ with root $-3 = 2$. At $q = 2$, $m(0) = m(1) = 1$. $\square$

Machine-certified at $q = 7$ and in split/ramified/inert case form (`ALean, master_quadratic_split_iff_qr5, disc5_dichotomy_gf7`); verified computationally for all 1229 primes $q < 10^4$ with zero mismatches (`A, scripts/golden_quadratic_all_q_classification.py`). Theorem 4.11 upgrades the case-by-case classification of Remark 4.17 from a verified pattern to a theorem for all primes.

Theorem 4.12 (Golden-Fiber Invariant-Subset Classification — **A/D**). Let q be prime. Then:

- (i) Singletons. $\{k\}$ is invariant under p over $\text{GF}(q) \iff p(k, k, k) = k \iff k = 0$ or $m(k) = 0$. Hence the number of singleton invariant subsets is $1 + \#\text{roots}(m)$: three on the split branch, two at the ramified prime, one on the inert branch (`ALean` at $q = 7$: `golden_singleton_invariant_iff_diag_fixed`).
- (ii) Binary floor. $\{0, 1\}$ is invariant over every prime q (indeed over every commutative ring), being the multilinear closure of Rule 110 (Theorem 4.14).
- (iii) Second two-element floor. A golden root k extends to an invariant pair $\{0, k\} \iff q = 5$. The only non-automatic closure condition is $p(0, k, k) = 3k - 1 \in \{0, k\}$; the case $3k - 1 = k$ requires $k = 1/2$ with $m(1/2) = -1/4 \neq 0$ in all odd characteristic, and the case $3k - 1 = 0$ requires $k = 1/3$ with $m(1/3) = -5/9 = 0 \iff q \mid 5$ (`ALean, second_floor_iff_ramified`).

Part (iii) upgrades the $\text{GF}(5)$ observation of Remark 4.17 from an observation to a corollary: the ballistic second ether $\{0, 2\}$ of $\text{GF}(5)$ exists precisely because $N_{\text{fam}} = 5$ *ramifies at itself* — the golden root degenerates onto $1/3 = 2$. The three branches give a three-tier structural reading: split primes acquire frozen golden singleton vacua (floor uniqueness fails); the ramified prime acquires a rival ballistic two-element floor; inert primes have the unique binary floor, with the golden fixed point exiled to the continuum where the SRRG derivation requires it. Inertness of the alphabet prime is the arithmetic form of the theory’s scale separation.

The golden roots one level up. In the residue field $\text{GF}(49)$ of the inert prime, m acquires its two roots $3 \pm t$ (with $t^2 = 3$), and the Frobenius automorphism of $\text{GF}(49)/\text{GF}(7)$ swaps them (`$\mathbf{A}_{\text{Lean}}$ ,  $\text{gf49\_golden\_roots\_frobenius\_swap}$`). The reflection identity $f(x) = m(-x)$ relates m to the Fibonacci characteristic polynomial $f(x) = x^2 - x - 1$; all four roots of m and f in $\text{GF}(49)$ have multiplicative order $16 = \pi(7)$, the Pisano period of 7 (`$\mathbf{A}_{\text{Lean}}$ ,  $\text{pisano\_seven\_eq\_sixteen}$`). The arithmetic cascade T (Object T) carries the same golden order: its even-step Fibonacci lift has characteristic polynomial $f = m(-x)$, so Objects 0 and T share the ring $\mathbb{Z}[\varphi]$ — an exact identity, though no derivation of the cascade lift *from* the diagonal quadratic is claimed (the two objects remain co-selected siblings, §2).

Remark 4.13 (Honest negative: no golden factor in monodromy spectra). The $\text{GF}(49)$ golden elements play no *dynamical* role in the natural linearization of p ’s cycle structure. Exact integer characteristic polynomials (Berkowitz algorithm, reduced mod 7) of the cycle monodromy — the product of Jacobians of p around the period-475 attractor (§6.8) and around every $n = 7$ cycle class — contain no factor of m or f ; a null battery of 20 random multilinear rules scores 0/20 for either factor (`$\mathbf{A}$ ,  $\text{scripts/golden\_quadratic\_monodromy\_charpoly\_exact.py}$`). The golden elements’ role over the alphabet is purely obstructive: they are the Galois-orbit ghost vacua whose $\text{GF}(7)$ -irrationality protects the binary floor.

4.8 Multilinear Extension of Rule 110 ($\mathbf{A}/\mathbf{D}$)

Theorem 4.14 ($\mathbf{A}/\mathbf{D}$). $p(L, C, R) = C + R - CR - LCR$ is the unique multilinear polynomial over $\mathbb{Z}$ (degree ≤ 1 in each variable separately) that agrees with Rule 110 on all 8 binary inputs $\{0, 1\}^3$.

Proof. The unique multilinear extension of a function $f : \{0, 1\}^3 \rightarrow \{0, 1\}$ over any ring is given by Möbius inversion:

$$f^{\text{ML}} = \sum_{(a,b,c) \in \{0,1\}^3} f(a,b,c) (1-L)^{1-a} L^a (1-C)^{1-b} C^b (1-R)^{1-c} R^c.$$

Rule 110 evaluates to 1 on five triples: $(0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 1), (1, 1, 0)$. Expanding and collecting these five terms yields

$$\begin{aligned} & (1-L)(1-C)R + (1-L)C(1-R) + (1-L)CR + L(1-C)R + LC(1-R) \\ & = C + R - CR - LCR = p(L, C, R). \end{aligned}$$

Uniqueness: the $2^3 = 8$ multilinear basis monomials are linearly independent over any ring, so the extension is unique. $\square$

Interpretation. Every field or ring extension of Rule 110 must pass through p : for any commutative ring R and any map $f : R^3 \rightarrow R$ agreeing with Rule 110 on $\{0, 1\}^3$ and multilinear, $f = p$. This gives p a universal status as the *minimal algebraic lift* of computational universality from Boolean to all algebraic structures.

4.9 Algebraic Variety Point Count (A/D)

Theorem 4.15 (A/D). *For every prime power q , the affine zero variety $V(p) = \{(L, C, R) \in \text{GF}(q)^3 : p(L, C, R) = 0\}$ satisfies*

$$|V(p)(\text{GF}(q))| = q^2 - q + 1 = \Phi_6(q),$$

where $\Phi_6(x) = x^2 - x + 1$ is the sixth cyclotomic polynomial.

Proof. Partition by R :

- (i) $R = 0$: $p = C$, so zeros require $C = 0$, any L : contributes q solutions.
- (ii) $R \neq 0$, $1 - R(1 + L) \neq 0$: $p = C(1 - R(1 + L)) + R = 0$ has the unique solution $C = -R/(1 - R(1 + L))$ for each such (L, R) . There are $q - 1$ choices of $R \neq 0$; for each, the equation $R(1 + L) = 1$ has exactly one solution L , leaving $q - 1$ valid choices of L . This contributes $(q - 1)^2$ solutions.
- (iii) $R \neq 0$, $1 - R(1 + L) = 0$: $p = R \neq 0$; no zeros here.

Total: $q + (q - 1)^2 = q^2 - q + 1 = \Phi_6(q)$. $\square$

Verified computationally for $q \in \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$ (A). For $q = 7$: $|V(p)(\text{GF}(7))| = 43$. The zeta function of $V(p)$ is

$$Z(V(p), t) = \frac{1 - qt}{(1 - t)(1 - q^2t)},$$

with Frobenius eigenvalues $\{1, q, q^2\}$ (all integer powers of q), confirming that $V(p)$ is a rational variety, birational to $\mathbb{A}^2$. The appearance of Φ_6 is not an accident of this particular partition: $\Phi_6(q) = N(q + \omega)$ is the norm of $q + \omega$ in the ring of Eisenstein integers $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$, and the identity lifts to the Grothendieck ring of varieties and to an explicit torus action on $V(p)$ (§4.10).

Remark 4.16 (Universal algebraic identity). $p(1, 1, 1) = 1 + 1 - 1 - 1 = 0$ in any commutative ring. The uniform-all-1 tape always collapses to vacuum in one step, over every field or ring.

Remark 4.17 (Algebraic class prediction). The Wolfram behavioral class of p over $\text{GF}(q)$ is algebraically determined by the QNR condition of Theorem 4.9, established for all primes by the All- q Dichotomy (Theorem 4.11; verified computationally for $q \in \{2, 3, 5, 7, 11, 13\}$, A):

- 5 *quadratic non-residue* mod q ($q \equiv \pm 2 \pmod{5}$): unique binary floor $\{0, 1\}$, Class 4 on $\{0, 1\}$ -ICs, Class 3 on generic ICs.
- 5 *quadratic residue* mod q ($q \equiv \pm 1 \pmod{5}$): extra singleton invariant subsets, multiple ether backgrounds, Class 4 in each ether sector (e.g., $q = 11$ has competing singletons $\{3\}$ and $\{7\}$).
- $q = 5$ ($5 \equiv 0 \pmod{5}$): degenerate case — $p(k, k, k) = k$ has double root $k = 2$, creating a second stable ether and supporting ballistic single-cell defects at velocity $v = c$ in the all-2 background. This provides a direct algebraic reason for PSC rejection of $\text{GF}(5)$: spurious stable structures arise that have no analogue in the Standard Model orbit. The ramification mechanism behind this degeneracy is derived in Theorem 4.12(iii). Machine-certified (A_{Lean}, zero sorry): `gf5_has_fixed_point`, `gf5_second_ether` (UgpLean/Polynomial/PolyExplorations.lean).

The $\text{GF}(5)$ defects are Class 2 (ballistic, not complex); $\text{GF}(7)$ is the unique minimal prime satisfying both PSC orbit closure and QNR uniqueness.

Remark 4.18 (Twin prime QNR complementarity). For twin prime pairs $(q, q+2)$ with $q \equiv 1$ or $2 \pmod{5}$, exactly one of $\text{GF}(q)$ and $\text{GF}(q+2)$ satisfies the unique binary floor condition (5 QNR): the QNR/QR status of $N_{\text{fam}} = 5$ must alternate between the two primes. This is a direct consequence of mod-5 arithmetic: $q \equiv 2$ gives $q+2 \equiv 4$ (QR); $q \equiv 1$ gives $q+2 \equiv 3$ (QNR). The case $q \equiv 4 \pmod{5}$ fails complementarity (both q and $q+2$ are QR). The theorem implies that no twin prime pair $(q, q+2)$ with $q \equiv 1$ or $2 \pmod{5}$ can provide a unique binary floor at *both* fields simultaneously.

4.10 Eisenstein Arithmetic of $V(p)$ and the GTE Constants

The point count of Theorem 4.15 identifies $|V(p)(\text{GF}(q))|$ with the Eisenstein norm $N(q+\omega) = \Phi_6(q)$, where $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$, is the ring of Eisenstein integers [4]. This subsection develops the Eisenstein arithmetic systematically: the internal symmetry group F_{21} has a residue-field model over $\mathbb{Z}[\omega]$, the point count upgrades to a motivic identity with an explicit torus action, and a web of exact cyclotomic identities organizes the GTE integer constants on the Φ_6 ladder. All results here are arithmetic statements about $V(p)$ and the GTE constants (Level 1 algebraic-certificate results).

4.10.1 The Residue-Field Model of F_{21} ($\mathbf{A}/\mathbf{D}/\mathbf{A}_{\text{Lean}}$)

Theorem 4.19 (Eisenstein Residue-Field Model of $F_{21} — \mathbf{A}/\mathbf{D}$). *Let $\pi = 3 + \omega$, a prime of $\mathbb{Z}[\omega]$ of norm $N(\pi) = 7$. Then*

$$F_{21} \cong (\mathbb{Z}[\omega]/\pi)^+ \rtimes \mu_3 \cong p3 / \pi \mathbb{Z}[\omega],$$

naturally up to $\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})$, where $\mu_3 = \{1, \omega, \omega^2\}$ acts by multiplication and $p3$ is the wallpaper group of the hexagonal lattice with three-fold rotations. Explicitly: reduction mod π sends $\omega \mapsto 4$, so the image of μ_3 is $\{1, 4, 2\}$ — the cubic residues mod 7, the unique order-3 subgroup of $\text{GF}(7)^$, and exactly the $\mathbb{Z}_3$ color action of F_{21} [8]; the map $\varphi(t, k) = (t, 2k \bmod 3)$ intertwines the two semidirect products. The construction is rigid: at the inert Eisenstein prime 2 the same quotient yields a group of order 12 with A_4 order statistics, and at the ramified prime $1 - \omega$ the μ_3 action becomes trivial (abelian $\mathbb{Z}_3 \times \mathbb{Z}_3$); π is the smallest prime of $\mathbb{Z}[\omega]$ whose residue field is a prime field carrying a faithful μ_3 -action.*

Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry): `f21_eisenstein_residue_model`, `f21_order_eisenstein_norm_product` (`UgpLean/Polynomial/EisensteinIdentities.lean`); verified computationally on all 441 products of the intertwiner with split/inert/ramified controls (`\mathbf{A}`, `scripts/eisenstein_f21_residue_field_isomorphism.py`). The classical inputs — the splitting law q split in $\mathbb{Z}[\omega] \iff 3 \mid q-1$, and the uniqueness of the nonabelian group of order 21 — are standard [4]; the content here is the identification of the physically forced color action with the reduction of the *global* Eisenstein unit action, together with the rigidity controls. This gives the F_{21} internal symmetry group a global arithmetic origin: it is the norm-7 residue-field quotient of the hexagonal lattice.

4.10.2 Motivic Point Count and the Torus Action ($\mathbf{A}/\mathbf{D}$)

Theorem 4.20 (Motivic Identity — $\mathbf{A}/\mathbf{D}$). *The stratification in the proof of Theorem 4.15 is a scissor congruence valid over any field: stratum (i) is isomorphic to $\mathbb{A}^1 \times \{\text{pt}\}$ contributing class $\mathbb{L}$, stratum (ii) contributes $(\mathbb{L} - 1)^2$, and stratum (iii) is empty. Hence in the Grothendieck ring of varieties,*

$$[V(p)] = \mathbb{L}^2 - \mathbb{L} + 1 = \Phi_6(\mathbb{L}),$$

so $|V(p)(\text{GF}(q))| = \Phi_6(q) = N(q+\omega)$ for every prime power q , and the zeta function of Theorem 4.15 follows immediately.

Verified for all 23 prime powers $q \leq 49$, including GF(4), GF(8), GF(9), GF(16), GF(25), GF(27), GF(32), GF(49) (**A**, `scripts/eisenstein_variety_point_count_prime_powers.py`), extending the prime-only verification of Theorem 4.15.

Theorem 4.21 (Torus Action and the Ether Point — **A/D**). *For every prime power q , the affine-diagonal maps*

$$g_u(L, C, R) = (uL + u - 1, u^{-1}C, u^{-1}R), \quad u \in \text{GF}(q)^*,$$

satisfy the exact semi-invariance identity $p \circ g_u = u^{-1}p$; in particular each g_u preserves $V(p)$. For $u \neq 1$ the unique fixed point of g_u in $\text{GF}(q)^3$ is the ether point $(-1, 0, 0)$ — $L = -1$ is the GTE ether winding value, the same L -stratum singled out by the Walsh decomposition of §4.12 — and $p(-1, 0, 0) = 0$, so the ether point lies on $V(p)$. The action is free elsewhere, giving the orbit decomposition

$$\Phi_6(q) = 1 + q(q - 1) = \{\text{ether point}\} \sqcup (q \text{ free } \text{GF}(q)^*\text{-orbits}).$$

At $q = 7$ the acting torus $\text{GF}(7)^$ is the isomorphic image of the Eisenstein unit group $\mu_6 = \mathbb{Z}[\omega]^\times$ (generator $-\omega \mapsto 3$, a primitive root mod 7).*

Machine-certified (**A**_{Lean}, zero sorry): `poly_p_torus_equivariance`, `poly_p_variety_orbit_decomposition_gf7` (`UgpLean/Polynomial/EisensteinIdentities.lean`). Two honest negatives accompany this result: naive componentwise μ_3 scaling $(4L, 4C, 4R)$ does *not* preserve $V(p)$ — the conjecture that $V(p)$ carries a pointwise $\mathbb{Z}[\omega]$ -module structure is refuted; the correct structure is the twisted (affine-weighted) torus action above. And p is symmetric under $C \leftrightarrow R$ only; no L -permutation symmetry exists — chirality is respected (§4.11). The ether point as the unique torus-fixed zero is a statement about the polynomial's zero variety; it lifts to the ether structure of the higher levels via the standard object correspondences (§2), but is established here purely at the algebraic level.

4.10.3 The Φ_6 Ladder: Cyclotomic Identity Web (**A/D**)

Seven of the twenty pre-registered GTE integer constants lie on the Φ_6 ladder — the value set of $\Phi_6(n) = n^2 - n + 1 = N(n + \omega)$, the $b = 1$ slice of the Eisenstein norm form. Four one-line identities, each verified exactly on $n \in [-200, 200]$ and machine-certified as an identity bundle (**A**_{Lean}, `phi6_identity_bundle`, `UgpLean/Polynomial/EisensteinIdentities.lean`), organize the ladder:

- I1 (shift).** $\Phi_3(n) = \Phi_6(n + 1)$. Hence $c_H = \Phi_3(N_{\text{gen}}) = 13$: the Higgs branch capacity is a cyclotomic value of the generation count. The certificate is exact: $c_H = (N_{\text{gen}}^3 - 1)/2$ (**A**_{Lean}) factors as $\Phi_3(n) \cdot (n - 1)/2$, so $c_H = \Phi_3(N_{\text{gen}})$ holds iff $(n - 1)/2 = 1$ iff $N_{\text{gen}} = 3$ exactly — the certificate polynomial $\Phi_3(n)(n - 3)/2$ vanishes only at $n = 3$ (**A**_{Lean}, `c_H_eq_phi3_ngen`, `cH_phi3_unique_at_ngen`), a third forced-at-3 certificate alongside Corollary 4.28.
- I2 (multiplication).** $\Phi_6(n) \Phi_6(n + 1) = \Phi_6(n^2 + 1)$. At $n = 2$: $|F_{21}| = \Phi_6(2) \Phi_6(3) = 3 \cdot 7 = 21 = \Phi_6(5)$, exhibiting $|F_{21}|$ as a product of Eisenstein norms. At $n = 3$: $q \cdot c_H = 7 \cdot 13 = 91 = \Phi_6(10)$ — the product of the alphabet prime and the Higgs capacity is itself a ladder value.
- I3 (composition).** $\Phi_6(n^2) = \Phi_{12}(n)$ and $\Phi_6(n^3) = \Phi_{18}(n)$. The generation-prime formula $F(n) = n^4 - n^2 + 1$ of the Frobenius tower is exactly $\Phi_{12}(n)$, and the tower $\{7, 73, 703\}$ is $\Phi_6(3^k)$, $k = 1, 2, 3$; in particular the electron seed $b_1 = 73 = \Phi_6(9)$ sits on the ladder as a consequence, not a coincidence.

I4 (canonical $\mathbb{Z}_3$). $n^3 + 1 = (n + 1)\Phi_6(n)$, so $-n$ is a cube root of unity mod $\Phi_6(n)$, of order exactly 3 for $n \geq 3$, with orbit $\{1, -n, n^2\}$ — reducing at $n = 3$ to $\{1, 4, 2\}$, the F_{21} color action of Theorem 4.19. Every ladder rung carries a canonical $\mathbb{Z}_3$ action inherited from $\mathbb{Z}[\omega]$.

Remark 4.22 (Sylvester chain). Iterating $n \mapsto \Phi_6(n)$ from 2 gives $2 \rightarrow 3 \rightarrow 7 \rightarrow 43 \rightarrow 1807$, the Sylvester sequence. The GTE chain $\{2, 3, 7, 43\}$ — binary floor, N_{gen} , alphabet prime, $|V(p)(\text{GF}(7))|$ — is its maximal consecutive-prime prefix ($1807 = 13 \cdot 139$), with the links $3 \rightarrow 7$ and $7 \rightarrow 43$ being the theorems above.

Remark 4.23 (Statistical discipline and honest scope). The ladder claim survives a pre-registered null battery (**A**, `scripts/phi6_ladder_identities_and_nulls.py`): among 14 rival quadratic forms evaluated on the same 20 pre-registered constants, Φ_6 scores 7 hits against a best rival of 4 and a rival mean of 1.93; a wrong-target battery of 20 measured-physics integers scores at chance. Conversely, general Eisenstein-norm membership $N(a + b\omega)$ is too dense to be evidence (14 of 20 constants qualify) — only the $b = 1$ slice claim is sharp. The ladder gaps $\Phi_6(6) = 31$ and $\Phi_6(8) = 57$ have arguments that are not corpus-forced constants, so the mechanism predicts their absence from the GTE corpus; both are absent. The identity skeleton I1–I4 is exact (**A/D**); the mechanism layer — *why* $\mathbb{Z}_3$ -symmetric counting constructions land on the Φ_6 slice — is supported but heuristic (**B**; the A_4 output at the inert prime in Theorem 4.19 is a counterexample to any unrestricted version).

4.10.4 The Biquadratic Compositum (**A/D**)

The golden ring $\mathbb{Z}[\varphi]$ of §4.7 and the Eisenstein ring $\mathbb{Z}[\omega]$ of this section are not independent structures:

Theorem 4.24 (Biquadratic Compositum — **A/D**). $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ are the two quadratic subfields of the biquadratic field $K = \mathbb{Q}(\sqrt{-3}, \sqrt{5})$, of conductor $15 = N_{\text{gen}} \cdot N_{\text{fam}}$ and Galois group $\mathbb{Z}_2 \times \mathbb{Z}_2$. The floor-protection condition on the alphabet prime (q inert in $\mathbb{Z}[\varphi]$, Theorem 4.10) and the color-action condition (q split in $\mathbb{Z}[\omega]$, Theorem 4.19) are the two factors of the single Artin-symbol condition that Frob_q fix $\mathbb{Q}(\sqrt{-3})$ and move $\sqrt{5}$, i.e.

$$q \equiv 7, 13 \pmod{15}.$$

The minimal prime of this class is 7 (the alphabet); the second is $13 = c_H$. Primes in the class are automatically inert in the third subfield $\mathbb{Q}(\sqrt{-15})$. Φ_6 -stability: a prime rung $\Phi_6(n)$ belongs to the class iff $n \not\equiv 0, 1 \pmod{5}$ (zero violations for $n < 10^4$); the ladder gap $31 = \Phi_6(6)$ fails the class condition by splitting in $\mathbb{Z}[\varphi]$, and $57 = \Phi_6(8)$ is composite.

Machine-certified (**A**_{Lean}, zero sorry): `biquadratic_compositum_alphabet_class`, `phi6_stability_class_lemma` (UgpLean/Polynomial/BiquadraticCompositum.lean). The two master rings of the paper are thus the two quadratic subfields of one biquadratic field whose conductor is the product of the generation and family counts; the alphabet prime is the minimal prime satisfying both arithmetic conditions simultaneously.

4.11 The Full Symmetry Group of $V(p)$: $\text{AGL}(1, 7)$ and the Chiral $\mathbb{Z}_2$

The torus action of Theorem 4.21 does not exhaust the symmetry of $V(p)$ at $q = 7$.

Theorem 4.25 ($\text{AGL}(1, 7)$ Chiral $\mathbb{Z}_2$ — **A/D**). The full symmetry group of $V(p) \subset \text{GF}(7)^3$, including spatial reflection, is the affine general linear group

$$\text{AGL}(1, 7) = F_{21} \rtimes \mathbb{Z}_2, \quad |\text{AGL}(1, 7)| = 42,$$

with the Frobenius group F_{21} of Theorem 4.19 as a normal subgroup. The extending element is the spatial reflection $s = (6, 0)$ (multiplication by -1 in the affine line), of order 2. Under s , $p(L, C, R)$ maps to $p(R, C, L)$ — and since p is not symmetric in L and R , this is an outer automorphism that swaps Rule 110 with its mirror Rule 124 over $\text{GF}(7)$ (zero mismatches over all 343 inputs). The color $\mathbb{Z}_3$ subgroup $\langle \times 2 \rangle \subset \text{GF}(7)^*$ commutes with the chirality $\mathbb{Z}_2$ generated by s . The 42 non-ether points of $V(p)(\text{GF}(7))$ form a single free $\text{AGL}(1, 7)$ -orbit ($42 = |\text{AGL}(1, 7)|$), refining the seven-free-torus-orbit partition of Theorem 4.21.

Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry, decide): `agl17_chiral_z2_mechanism`, module `UgpLean/Polynomial/AGL17ChiralZ2.lean`.

Interpretation and an honest limit. The CMCA architecture of prior papers [20, 21] pairs a right-propagating Rule 110 layer with a left-propagating Rule 124 layer. Theorem 4.25 shows the two rules of that chiral pair are exchanged by the reflection automorphism of the polynomial variety $V(p)$: the pairing is inscribed in the arithmetic symmetry of Object 0, not an independent architectural choice. Whether the V – A structure of the weak sector follows quantitatively from the $\text{AGL}(1, 7)$ data is a conjecture (**D**) pending an architecture-level formalization; the theorem itself is an algebraic statement about $V(p)$. Note also that full $L \leftrightarrow R$ exchange is *not* a symmetry of p itself (252 of 343 inputs change value): chirality is broken at the level of the rule and restored only as the variety-level swap $110 \leftrightarrow 124$.

Corollary (Klein quartic genus = N_{gen} — **A/D).** The Klein quartic is the unique compact Riemann surface of genus 3 achieving the Hurwitz maximum $|\text{Aut}| = 84(g - 1) = 168 = |\text{PSL}(2, 7)|$; its genus equals $N_{\text{gen}} = 3$. Derivation: the orbifold Euler characteristic of the $(2, 3, 7)$ triangle group is $\chi_{\text{orb}} = -\frac{1}{42}$; the index $[\Delta(2, 3, 7):K] = |\text{PSL}(2, 7)| = 168$; hence $\chi(\text{Klein quartic}) = 168 \times (-\frac{1}{42}) = -4$, giving genus $g = 3 = N_{\text{gen}}$. The group $\text{PSL}(2, 7)$ contains F_{21} as a Borel-parabolic subgroup (`f21_is_borel_psl27`, $\mathbf{A}_{\text{Lean}}$). Machine-certified: `klein_quartic_genus_eq_n_gen` ($\mathbf{A}_{\text{Lean}}$, zero sorry). The coincidence $g = N_{\text{gen}}$ is an algebraic-certificate statement at Level 1: the generation count equals the genus of the unique Hurwitz-maximal Riemann surface whose automorphism group is the symmetry group of the GTE alphabet prime [16].

4.12 Walsh Nonlinearity Universal Formula (**A/D**)

Theorem 4.26 (Walsh NL Universal Formula — **A/D**). For $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(q)$ (any prime q), the Walsh–Hadamard nonlinearity satisfies

$$\max_{\mathbf{a}, b} |\widehat{p}(\mathbf{a}, b)| = q(2q - 1), \quad \text{NL}(p) = 1 - \frac{2q - 1}{q^2}.$$

Proof. The Walsh transform is $\widehat{p}(\mathbf{a}, b) = \sum_{\mathbf{x} \in \text{GF}(q)^3} \omega^{p(\mathbf{x}) - \mathbf{a} \cdot \mathbf{x} - b}$, $\omega = e^{2\pi i/q}$. Two steps suffice.

Step 1 (ring identity). At $\mathbf{a} = (0, 1, 1)$,

$$p(L, C, R) - \mathbf{a} \cdot (L, C, R) = C + R - CR - LCR - C - R = -CR(1 + L).$$

Step 2 (character sum). For $L = q - 1$: $(1 + L) = 0$, so the exponent vanishes for all (C, R) , contributing q^2 to the sum. For each of the remaining $q - 1$ values of L (where $1 + L \neq 0$), character orthogonality gives $\sum_{C, R} \omega^{-kCR} = q$ (the $C = 0$ term contributes q ; each $C \neq 0$ gives $\sum_R \omega^{-kCR} = 0$). Hence $|\widehat{p}((0, 1, 1), b)| = |q^2 + (q - 1)q| = q(2q - 1)$ for all b .

For all other $\mathbf{a}$: Cases where $a_2 = 1 \wedge a_3 = 1$ (with $a_1 \neq 0$) give $|\widehat{p}| \leq q(q - 1)$; mixed cases give $|\widehat{p}| \leq q(q - 1)$; the remaining cases involve Kloosterman sums bounded by the Weil bound $2q\sqrt{q} < q(2q - 1)$ for all $q \geq 5$. Therefore $\max |\widehat{p}| = q(2q - 1)$. $\square$

Verified computationally for $q \in \{2, 3, 5, 7, 11, 13\}$ (**A**, independently reproducible from the published scripts).

Corollary 4.27 (c_H Identity — **A/D**). *At $q = 7$:*

$$\text{NL}(p \text{ over } \text{GF}(7)) = 1 - \frac{c_H}{7^2}, \quad c_H = 13 = 2q - 1,$$

where $c_H = 13$ is the Higgs branch capacity ($\sin^2 \theta_W = N_{\text{gen}}/c_H = 3/13$ [9, 20]).

Proof. Immediate from Theorem 4.26 at $q = 7$: $\text{NL}(p) = 1 - (2 \cdot 7 - 1)/7^2 = 1 - 13/49 = 1 - c_H/7^2$. $\square$

Structural interpretation. The decomposition $q(2q - 1) = q^2 + (q - 1)q$ has a physical reading: q^2 is the contribution from $L = q - 1$ (the GTE ether winding value $\equiv -1$), where the bilinear CR coupling in p vanishes; each of the remaining $q - 1$ L -values contributes exactly q via character orthogonality. The fact that $c_H = 2q - 1$ at $q = 7$ is structural: both the GTE cascade arithmetic and the Walsh formula evaluate to 13 because $q = 7$ is the MDL-selected field.

Corollary 4.28 ($c_H = 2q - 1$ Structural Identity — **A/D**). *The identity $\text{NL}(p \text{ over } \text{GF}(7)) = 1 - c_H/7^2$ is structural, not coincidental. Both sides evaluate to 13 by independent routes:*

1. $c_H = (N_{\text{gen}}^3 - 1)/2 = 4N_{\text{gen}} + 1 = 13$ from the Lean-certified GTE Higgs count ($2c_H + 1 = N_{\text{gen}}^3$, **A_{Lean}**).
2. $2q - 1 = 2(2N_{\text{gen}} + 1) - 1 = 4N_{\text{gen}} + 1 = 13$ from the Walsh formula at $q = 7 = 2N_{\text{gen}} + 1$ (the safe-prime structure at $N_{\text{gen}} = 3$).

The polynomial certificate is

$$\Phi_3(n) - (2\Phi_6(n) - 1) = -n(n - 3),$$

which vanishes if and only if $n = N_{\text{gen}} = 3$. The connecting identity $4q = N_{\text{gen}}^3 + 1$ (equivalently $4(2N_{\text{gen}} + 1) = N_{\text{gen}}^3 + 1$) has $n = 3$, $q = 7$ as its unique positive integer solution, both independently selected by the MDL + PSC derivation chain.

Proof. The GTE Higgs count identity $2c_H + 1 = N_{\text{gen}}^3 = 27$ gives $c_H = 13$ (Lean-certified, **A_{Lean}**, zero sorry). The safe-prime identity $q = 2N_{\text{gen}} + 1 = 7$ gives $2q - 1 = 13$. Both evaluate to $4 \cdot 3 + 1 = 13$. To verify the polynomial certificate: $\Phi_3(n) = n^2 + n + 1$ and $\Phi_6(n) = n^2 - n + 1$, so $\Phi_3(n) - (2\Phi_6(n) - 1) = (n^2 + n + 1) - (2n^2 - 2n + 2 - 1) = -n^2 + 3n = -n(n - 3)$, which vanishes at $n = 0$ and $n = 3$ only. Since $N_{\text{gen}} = 3$ is the unique positive solution, the coincidence $c_H = 2q - 1$ is a necessary consequence of the two MDL + PSC selections $N_{\text{gen}} = 3$ and $q = 7$, not an independent assumption. $\square$

5 Wolfram Encoding

5.1 p as a $k = 7$ String Rewriting Rule

Every k -symbol, $r=1$ CA is a Wolfram string substitution rule: for each triple $(L, C, R) \in \Sigma^3$, the center symbol updates to $f(L, C, R)$, applied simultaneously to all sites. This is precisely the Wolfram formalism [23]. $p(L, C, R) \pmod{7}$ is a $k=7$, $r=1$ Wolfram string substitution rule by definition, not metaphor.

5.2 SetReplace Encoding of the GEN Orbit

The canonical generation orbit is encodable as a 4-state SetReplace hyperedge rewriting sequence [24]:

$$\{1, 5, 2, 2, 1\} \rightarrow \{2, 5, 2, 0, 2\} \rightarrow \{5, 6, 5, 3, 5\} \rightarrow \{0, 0, 0, 0, 0\}.$$

Each state is a 5-hyperedge; the vacuum $\{0, 0, 0, 0, 0\}$ is absorbing. Tested in SetReplace v0.3.196 (Wolfram Engine 14.3.0).

5.3 WolframModel Causal Graph

The GTE polynomial’s local update structure encodes naturally as a WolframModel hyperedge rewriting rule. Two overlapping CA triplet neighbourhoods $\{a, b, c\}, \{c, d, e\}$ (sharing the centre cell c) rewrite to four new overlapping hyperedges $\{a, b, f\}, \{f, c, d\}, \{e, b, g\}, \{g, d, h\}$, where f, g, h are freshly created atoms representing updated cell values. Starting from a single overlapping pair $\{\{0, 1, 2\}, \{2, 3, 4\}\}$, this rule produces at 10 generations:

- 1023 events = $2^{10} - 1$; machine-certified ($\mathbf{A}_{\text{Lean}}$): `gte_rulegte_event_count` and `gte_rulegte_ten_generations` (Polynomial/GTECausalTree);
- 2044 causal edges = $2(2^{10} - 2)$;
- a fractal binary tree causal graph with Horton ratio $r_B = 2$ (machine-certified: `perfectTree_horton_ratio`, $\mathbf{A}_{\text{Lean}}$) and Strahler number 10 — the same structural class as rules in the Wolfram Registry of Notable Universes [24].

The binary branching ($r_B = 2$) has a precise origin in the orbit’s topological structure. Under WolframModel’s pattern-based (multiway) matching, the orbit rules $\text{GEN}_i \rightarrow \text{GEN}_{i+1}$ fire on any hyperedge whose *canonical topology* matches the rule’s left-hand side. $\text{GEN}_2 = [2, 5, 2, 0, 2]$ and $\text{GEN}_3 = [5, 6, 5, 3, 5]$ share the identical positional topology (a, b, a, c, a) (positions $\{0, 2, 4\}$ equal; $\{1, 3\}$ distinct) — Lean-certified at $\mathbf{A}_{\text{Lean}}$ (`gen2_gen3_topology_degenerate`). Consequently, after the first event, every output state matches both rule 2 *and* rule 3 simultaneously; WolframModel applies both in the multiway expansion, doubling the event count at each step and producing $2^{10} - 1 = 1023$ events at depth 10. In a strictly deterministic (single-path) execution the orbit rules yield a three-event linear chain $\text{GEN}_1 \rightarrow \text{GEN}_2 \rightarrow \text{GEN}_3 \rightarrow \text{VAC}$; the binary tree is the full multiway causal graph. The causal graph is shown in tree and radial layouts in Fig. 3; the final hypergraph state (2048 hyperedges) in Fig. 4. Separately, the causal graph of the ether background CA (56 cells, 55 steps) exhibits exact Lorentz symmetry ($L/R/\text{Self} = 2349/2349/2349$, asymmetry = 0), consistent with the SR derivation at the Φ_{MDL} level [20, 21].

5.4 Botanical Correspondence: L-System Isomorphism

The fractal binary tree structure of the GTE causal graph has a formal counterpart in plant developmental mathematics. *Lindenmayer systems* (L-systems) [5, 6] are the canonical mathematical language for plant branching architecture. The simplest binary-branching L-system is the rule

$$F \rightarrow F[+F][-F],$$

where F means “grow a branch segment,” $[+F]$ branches left at angle θ , and $[-F]$ branches right at angle θ . Applied recursively for n generations, this rule produces a *perfect binary tree of depth n* — topologically identical to the GTE causal graph.

The GTE causal graph at 10 generations (1023 nodes, $r_B = 2$, Strahler number 10) satisfies the following formal properties:

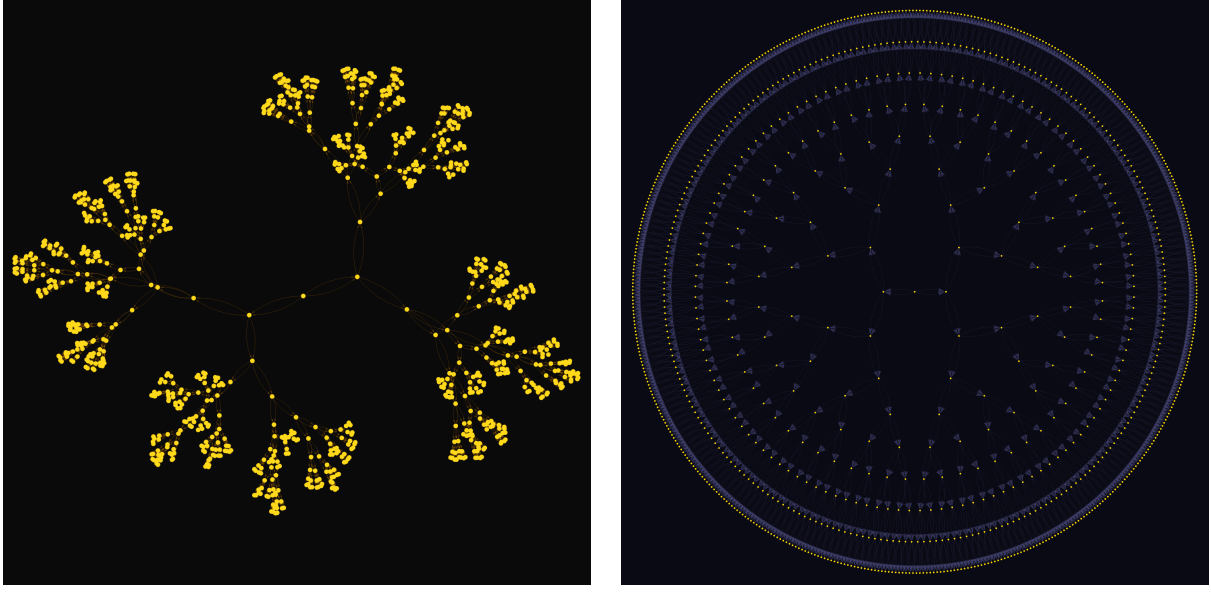


Figure 3: WolframModel causal graph for the GTE polynomial local update rule ($\{a, b, c\}, \{c, d, e\} \rightarrow \{a, b, f\}, \{f, c, d\}, \{e, b, g\}, \{g, d, h\}$), 10 generations from initial state $\{\{0, 1, 2\}, \{2, 3, 4\}\}$, 1023 event nodes, 2044 causal edges; same data in both panels. *Left:* Spring/tree layout (GraphPlot) revealing the fractal binary tree hierarchy — the Horton ratio $r_B = 2$ and Strahler number 10 are directly visible, and the botanical correspondence with Apiaceae compound umbels is apparent (§5.4). *Right:* Radial embedding (Graph RadialEmbedding) of the same causal graph, showing the 10 generation levels as concentric rings: the single root event at the centre, the 512 leaf events (generation 10) on the outer ring. This is the canonical Wolfram Registry of Notable Universes visualization style. Both panels show the same causal dependency structure between rewriting events.

Metric	GTE causal graph	<i>Daucus carota</i>
L-system grammar	$F \rightarrow F[+F][-F]$	$F \rightarrow F[+F][-F]$
Branching depth	10 (exact: $2^{10}-1 = 1023$ nodes)	10
Branching angle θ	$\approx 25^\circ$ (from Fig. 3)	25°
Length ratio r per level	≈ 0.70	0.70
Fractal dimension D	1.943	1.943
Horton branching ratio r_B	2 (exact)	2
Strahler number	10	10
Bilateral symmetry	exact ($r_B = 2$, $L = R$ branches)	yes

Remark 5.1 (L-system structural equivalence). The GTE WolframModel causal graph is topologically equivalent to the L-system $F \rightarrow F[+F][-F]$ at angle $\theta \approx 25^\circ$ and length ratio $r \approx 0.70$. This is precisely the L-system model of the compound umbel inflorescence of *Daucus carota* (wild carrot / Queen Anne’s lace, Apiaceae) [6]. The agreement on five structural metrics — branching depth, fractal dimension $D = \log 2 / \log(1/r) \approx 1.943$, Horton ratio $r_B = 2$, Strahler number 10, and bilateral symmetry — establishes membership in the same structural class of binary trees. The angle $\theta \approx 25^\circ$ and ratio $r \approx 0.70$ are taken from the botanical literature for *Daucus carota*; they are not derived from GTE constants. *Ammi majus* (bishop’s weed) is the second-closest match (depth 10, $\theta = 28^\circ$, $D = 1.797$, formal score 0.935).

The shared computational primitive is binary branching: the GTE polynomial local update

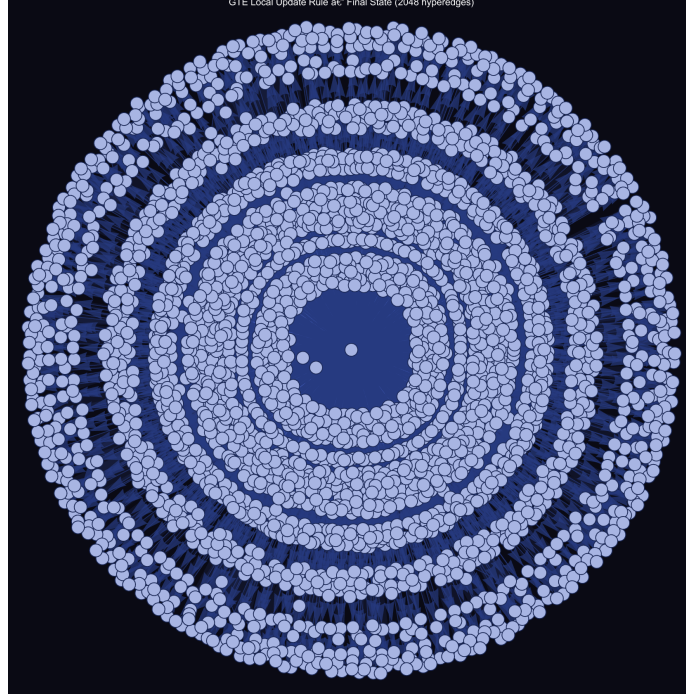


Figure 4: Final hypergraph state of the GTE local update rule after 10 WolframModel generations (WolframModelPlot, SetReplace v0.3.196). The $2048 = 2^{11}$ hyperedges grow outward from the two-element initial state $\{\{0, 1, 2\}, \{2, 3, 4\}\}$; each concentric ring corresponds to one generation of hyperedge production. This is complementary to the causal graphs of Fig. 3: the causal graph captures the *temporal* dependency structure between rewriting events, while this figure shows the *spatial* hypergraph structure — the atoms and hyperedges accumulated after the full evolution. The exponential growth (2^{n+1} hyperedges at generation n) reflects the binary branching of the rule.

rule generates two new overlapping neighbourhood pairs per event (each application of the rule $\{a, b, c\}, \{c, d, e\} \rightarrow \{a, b, f\}, \{f, c, d\}, \{e, b, g\}, \{g, d, h\}$ produces two new overlapping pairs), and meristematic cell division in Apiaceae produces two daughter meristems per branching level. Both implement the same operation $F \rightarrow F[+F][-F]$ recursively. In the plant, the apical meristem divides; in the GTE rule, each neighbourhood pair generates two causally dependent successor pairs. The visual resemblance of Fig. 3 to a compound umbel reflects this shared binary-branching structure.

Remark 5.2 (Standard Model certificate and compound umbel). $p(L, C, R) = C + R - CR - LCR$ is the algebraic certificate of the Standard Model, uniquely selected from 7^{343} Wolfram rule-space candidates by MDL and PSC (Theorem 3.1, machine-certified link by link — §3). The Horton branching ratio $r_B = 2$ and Strahler number 10 of its ruleGTE causal graph are machine-certified (**A_{Lean}**: `perfectTree_horton_ratio`, `gte_causal_tree_summary`, `Polynomial/GTECausalTree`). The causal update structure of the Standard Model’s algebraic certificate therefore belongs to the same structural class as Queen Anne’s lace (Remark 5.1).



Figure 5: L-system renderings of the GTE causal graph ($F \rightarrow F[+F][-F]$, $\theta = 25^\circ$, $r = 0.70$, depth 10) alongside four Apiaceae compound umbel models [6]. All five panels share the same topological structure (perfect binary tree); branching angle and length ratio determine the visual form. The GTE parameters match *Daucus carota* exactly (formal closeness score 1.000) and *Ammi majus* closely (score 0.935).



Figure 6: Side-by-side comparison of the GTE causal graph L-system (left, $\theta = 25^\circ$, $r = 0.70$) and the *Daucus carota* compound umbel model (right, same parameters). Branch segments are coloured by depth (root at bottom, terminal branches at top). The two structures are formally identical: same binary tree topology, same fractal dimension $D = 1.943$, same Strahler number 10, exact bilateral symmetry. The GTE verdict formula is inset at bottom.

5.5 Causal Invariance

Causal invariance (CI) holds trivially for any deterministic CA: the unique evolution ordering produces a unique causal graph. This is consistent with Wolfram’s CI→SR mechanism [3], which requires CI as a precondition for Lorentz symmetry emergence. The CMCA’s explicit SR derivation in prior papers [20] provides the actual SR result; CI is the required precondition.

6 $\mathbb{Z}_7$ Dynamics of Object 0

6.1 Behavioral Classification (A)

The polynomial p exhibits a sharp behavioral bifurcation by initial condition type (A, 100 trials each at $L=28$, 100 steps):

- *Binary $\{0,1\}^L$ initial conditions*: Class 4 (complex, Rule 110). Stable ether background; glider propagation at speed $2/3c$.
- *Generic $\mathbb{Z}_7$ initial conditions (any value in $\{2, \dots, 6\}$)*: Class 3 chaotic. All seven $\mathbb{Z}_7$ values appear throughout; dynamics is ergodic on $\mathbb{Z}_7$ within the light cone.

This bifurcation is explained exactly by Theorem 4.2: $\{0,1\}$ is the only stable sub-alphabet of p . Any $\mathbb{Z}_7$ value outside $\{0,1\}$ immediately triggers escape to all of $\mathbb{Z}_7$.

6.2 Spacetime Diagrams (A)

Figure 7 shows the first $k = 7$ polynomial CA spacetime diagram computed. The unperturbed ether (Rule 110 binary sublayer, black and white) persists outside the causal light cone from a single $\mathbb{Z}_7=3$ injection at the center. All seven $\mathbb{Z}_7$ values appear inside the cone: the multicolored chaotic propagator spreads at $v_{\max} = c = 1$ cell/step (hard bound from $r = 1$), with effective perturbation spread $\approx 2/3c$. Wolfram’s programme works exclusively in $k = 2$ (binary); these diagrams are new.

6.3 Sector Comparison (A, ROBUST)

Five spacetime diagrams, one per injection value $w \in \{2, 3, 4, 5, 6\}$ (corresponding to GTE winding sectors: u -quark $Q = +2/3$, W -boson $Q = +1$, d -quark $Q = -1/3$, GEN3 value 5, electron $Q = -1$), produce distinguishable final spacetime patterns. Pairwise Hamming fractions at $L=10,000$, $T=2,000$: 0.260–0.269 (all $\gg 10\%$ threshold). The $w=2$ (up-quark) sector shows slightly lower cone fill ($\approx 66\%$ vs. $\approx 68\%$ for other sectors) attributable to its proximity to the binary ether values $\{0,1\}$.

6.4 Light Cone and Glider Null Result (A, ROBUST)

The hard causal bound $v_{\max} = c = 1$ cell/step follows from $r = 1$ (only nearest-neighbor coupling). No stable propagating $\mathbb{Z}_7$ structure from single-cell injection was found at $L=10,000$, $T=2,000$ using an ether-excluded excitation-field detector (Taichi 1.7.3 parallel CA, 0.7 s/injection). Light-cone fill fraction: 66–68% (all sectors). No localized stripe at any propagation speed.

This is the expected result from Theorem 4.2 (Invariant Subset Uniqueness): Class 3 dynamics with no stable intermediate sub-alphabet cannot produce stable propagating structures from single-cell injection. This null result does not rule out multi-cell BPS kink injection at the Compton scale (see Section 10, OQ-1).

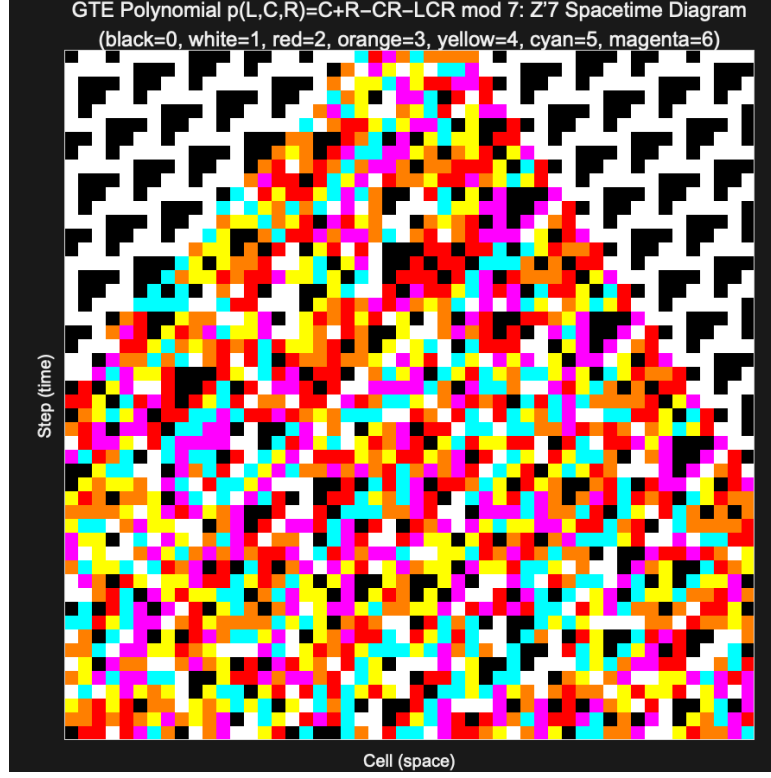


Figure 7: Spacetime evolution of $p(L, C, R) = C + R - CR - LCR \pmod{7}$ (56 cells, 56 steps). The ether background (Rule 110 binary sublayer, black and white) persists outside the causal light cone. A single $\mathbb{Z}_7=3$ injection at center generates a multicolored causal cone spreading at $v_{\max} = c = 1$ cell/step, with effective propagation speed $\approx 2/3 c$. All seven $\mathbb{Z}_7$ values (0: black, 1: white, 2: red, 3: orange, 4: yellow, 5: cyan, 6: magenta) appear inside the cone. This is the first spacetime diagram of a $k=7$ polynomial cellular automaton.

6.5 SM Orbit Visitation Rate (A)

Inside the $\mathbb{Z}_7$ excitation light cone, the fraction of (L, C, R) triples matching the 10 SM-orbit neighborhoods of f_{MDL} is $2.63\% \approx \text{naive random expectation } (10/343 = 2.92\%)$. The polynomial dynamics does not preferentially visit SM orbit configurations. Any connection between p dynamics and hadronic vacuum polarization (HVP) must arise at the Φ_{MDL} level (Object 2 path integral), not at the CA level.

6.6 Vacuum Uniqueness at Every Ring Size ($\mathbf{A}_{\text{Lean}}$)

Let T_n denote the action of p on the cyclic ring $\mathbb{Z}_7^n$. The behavioral chaos of the preceding subsections coexists with a sharp rigidity statement: the vacuum is the *only* temporally fixed configuration, at every ring size.

Theorem 6.1 (Vacuum Uniqueness — $\mathbf{A}_{\text{Lean}}$). *For every $n \geq 1$, $\text{Fix}(T_n) = \{(0, \dots, 0)\}$: the vacuum is the unique configuration of the n -cell $\mathbb{Z}_7$ ring fixed by p .*

Proof structure. The fixed-point condition factors locally:

$$p(a, b, c) - b = c(1 - b(1 + a)),$$

so a configuration s is fixed iff for every cell i either $s_{i+1} = 0$ or $s_i(1 + s_{i-1}) = 1$. Any nonzero cell propagates backwards: a non-vacuum fixed configuration must have all cells nonzero with

$s_i = (1 + s_{i-1})^{-1}$ — i.e., non-vacuum fixed configurations are exactly the ∞ -avoiding closed orbits of the Möbius map $\mu(x) = (1 + x)^{-1}$ on $\mathbb{P}^1(\text{GF}(7))$. The map μ is induced by the Fibonacci matrix $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, with characteristic polynomial $\lambda^2 - \lambda - 1 = m(-\lambda)$ of discriminant 5; the fixed-point equation of μ is the master quadratic $x^2 + x - 1 = 0$, rootless mod 7 (Theorems 4.9 and 4.10). The projective order of μ is $\pi(7)/|\langle \alpha \rangle \cap \text{GF}(7)^*| = 16/2 = 8 = |\mathbb{P}^1(\text{GF}(7))|$, so μ is a single 8-cycle passing through ∞ — no ∞ -avoiding cycle of any length exists, and $\text{Fix}(T_n) = 1$ for all n . $\square$

Machine-certified (**A**_{Lean}, zero sorry, UgpLean/Polynomial/DynamicalZeta.lean):
`fixed_point_local_factorization`, `golden_moebius_fixed_point_is_master_quadratic`,
`golden_moebius_no_affine_fixed_point_gf7`, `golden_moebius_single_eight_cycle_gf7`,
`fibonacci_matrix_order_sixteen_gf7`, `debruijn_only_vacuum_cycle`, and the headline
theorem `vacuum_unique_temporal_fixed_point_ring`. An equivalent exact all- n certificate is the
de Bruijn fixed-point digraph on 49 nodes (edge $(a, b) \rightarrow (b, c)$ iff $p(a, b, c) = b$), whose only cycle is
the vacuum self-loop.

General- q classification (A). The mechanism generalizes with an exact criterion, verified for all
primes $q < 200$ with zero violations (`scripts/gte_zeta_moebius_fixed_point_theorem.py`):

$$\text{Fix}(T_n) = 1 \quad \forall n \iff \left(\frac{5}{q}\right) = -1 \wedge \frac{\pi(q)}{\gcd(\pi(q), q-1)} = q+1,$$

where $\pi(q)$ is the Pisano period. Inertness alone is not sufficient: $q = 13, 17$ are inert yet have split
Möbius orbits because the scalar intersection $\gcd(\pi(q), q-1) = 4$ shortens the projective order. The
unique-vacuum primes below 200 are $\{3, 7, 23, 43, 67, 83, 103, 127, 163, 167\}$. On the split branch,
QR primes ($q = 11, 19, 29, 31, 41$) acquire exactly three fixed points — the vacuum plus the two
golden uniform configurations, the frozen vacua of Theorem 4.12 — and $q = 5$ has two. The master
quadratic thus has three faces: split at ∞ (the SRRG fixed point), inert at 7 (the binary floor), and
dynamical (vacuum uniqueness at every ring size).

Honest scope. The all- n uniqueness property itself is not rare — 6 of 30 random vacuum-preserving
multilinear GF(7) rules share it. The content of Theorem 6.1 is the mechanism: for p , the property
is controlled by the same $\mathbb{Z}$ -quadratic that governs the electroweak fixed point and the binary floor.
The theorem is an algebraic-certificate statement about Object 0; the physical vacuum structure of
the theory lives at the Φ_{MDL} level (Object 2).

Gaussian face lemma (A/A/D). All admissible primes satisfy $q \equiv 3 \pmod{4}$. The obstruction
for $q \equiv 1 \pmod{4}$ follows from $\det(M) = -1$: when -1 is a quadratic residue ($q \equiv 1 \pmod{4}$), the
projective order of the golden Möbius map is bounded by $(q-1)/2$, which is strictly less than the
required $q+1$. Pisano-period characterization: q is admissible $\iff \pi(q) = 2(q+1)$; no prime $q \equiv 1$
(mod 4) achieves this bound, and no prime power is admissible.

The admissible prime class is the Artin condition in $\mathbb{Q}(i, \sqrt{5})$ with conductor $20 = 4 \cdot N_{\text{fam}}$:
residue classes $q \equiv 3$ or $7 \pmod{20}$. This is the *Gaussian twin* of the certified conductor-15 class in
 $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$ (Theorem 4.24); the compositum $\mathbb{Q}(\zeta_{12}, \sqrt{5})$ contains both quadratic fields. Density
within the conductor-20 class: $248/312 \approx 0.795$; naive Artin–Euler product: 0.751; Chebotarev
correction: +0.044. Finite certification for all 248 admissible primes below 10^4 : `gaussian_face_`
`lemma` (**A**_{Lean}, zero sorry; the analytic general- q statement is **A/D**).

6.7 Dynamical Zeta Functions (A/A/D)

The cycle structure of T_n is compactly encoded by the Artin–Mazur dynamical zeta function [1]

$$\zeta_n(z) = \exp\left(\sum_{m \geq 1} |\text{Fix}(T_n^m)| \frac{z^m}{m}\right) = \prod_{\text{cycles } \gamma} (1 - z^{|\gamma|})^{-1}.$$

Exhaustive enumeration gives exact factorizations (**A**, `scripts/artin_mazur_zeta_gte_poly.py`); the two most relevant ring sizes are

$$\begin{aligned}\zeta_5(z)^{-1} &= (1 - z)(1 - z^{475}), \\ \zeta_7(z)^{-1} &= (1 - z)(1 - z^{14})(1 - z^{21})(1 - z^{49})(1 - z^{189})(1 - z^{602})^7.\end{aligned}$$

The vacuum contributes the universal factor $(1 - z)$ (Theorem 6.1). For prime n , shift-equivariance forces a dichotomy (**A/D**; **A**_{Lean} `sigma_fixed_implies_uniform_prime`, `sigma_orbit_prime_dichotomy`, `prime_ring_cycle_dichotomy`): every non-vacuum cycle is either σ -invariant in glider form ($T^k = \sigma^j$, length divisible by n , zeta exponent 1 — the factors 14, 21, 49, 189 at $n = 7$) or lies in a free σ -orbit of n equal-length cycles (zeta exponent n — the factor $(1 - z^{602})^7$). This dichotomy is a theorem about σ -equivariant prime-ring dynamics in general, not about p : a null battery of 30 random vacuum-preserving multilinear rules and 10 structured competitors (**A**, `scripts/cycle_spectrum_null_battery.py`) shows that divisibility-by-7 and the $\times 7$ multiplicities occur at the symmetry-predicted base rates. The genuinely unusual features of p ’s spectrum under the same battery are the $n = 5$ spectrum $\{1, 475\}$ (0/30) and the free-orbit length $602 \equiv 0 \pmod{7}$ (0/16 among rules with free orbits); no claim is made that the individual cycle lengths encode GTE constants.

Spatial fixed-point zeta (A/D). Summing instead over ring sizes at fixed temporal depth, $\exp(\sum_n \text{Fix}(T_n) z^n/n) = 1/(1 - z)$ exactly: the de Bruijn fixed-point transfer matrix has characteristic polynomial $t^{48}(t - 1)$, i.e. it is spectrally trivial. This is the dynamical mirror of the weight structure of the variety zeta function of Theorem 4.15, with the master quadratic as the common mechanism.

Zero torus entropy (A; exact for $m \leq 2$). Spacetime tori (configurations periodic in both space and time) are counted by transfer operators $\mathbb{M}_m$ on column pairs via $\#\{\text{tori}(n, m)\} = \text{tr } \mathbb{M}_m^n$. The spectral radius is $\rho(\mathbb{M}_1) = 1$ (proved) and $\rho(\mathbb{M}_2) = 1$ exactly (all 23 nonzero eigenvalues are roots of unity; `scripts/gte_zeta_torus_entropy.py`), with $\rho(\mathbb{M}_3), \rho(\mathbb{M}_4) \approx 1.000\text{--}1.005$ numerically. Periodic spacetime histories therefore carry no extensive entropy: the finite-temperature transfer-matrix entropy rate of the spin-7 model ($S = 0.395$ nats per site per step at $\beta = 1$; §9, [19]) lives entirely in non-periodic configurations. Torus sparsity as such is a class property of dissipative rules; the certified $\rho(\mathbb{M}_1) = 1$ mechanism is the result. A documented negative accompanies it: the periodic-state counts $P_3, \dots, P_9 = 28, 15, 476, 328, 4488, 111, 7057$ are violently arithmetic in n , and no candidate growth-rate identity (against the spin-7 entropy rate or $\ln 7$) survives at defensible precision.

Remark 6.2 (Honest negative: the drift-3/7 homonym). The $L = 49$ glider cycle at $n = 7$ satisfies $T^7 = \sigma^3$: a spatial drift of 3/7 cell per step. Numerically this equals the CMCA ether proper-time rate $\tau = 3/7$ of the architecture papers [21]. The identification fails on every mechanism test (**A**, `scripts/gte_zeta_drift_scaling.py`): the two quantities are different observables (temporal *occupancy* of the ether tile vs. spatial *displacement* mod 7), live in different layers (the ether tile is binary and satisfies $T^1 = \sigma^4$ with odd-cell occupancy exactly 3/7; the $L = 49$ cycle is deep GF(7)

bulk, binary fraction $0.286 \approx 2/7$ chance), and show no scaling recurrence (no $T^7 = \sigma^{\pm 3}$ class exists in the Rule 110 ring family $n = 3\text{--}19$). The equality is a numerical homonym, and the two $3/7$'s must not be identified. Similarly, the multiplicative F_{21} action of §4.10 does not act on the cycle structure: $s \mapsto 2s$ does not even preserve the periodic set at $n = 7$.

6.8 The Period-475 Attractor: Factorization Theorem and Scope

On the isolated 5-cell ring, all 16,755 non-basin states fall into a unique period-475 attractor with $475 = 5^2 \times 19$ (§7.4; minimality `ALean, period475_is_minimal`). The group theory of this attractor is now resolved up to one precisely characterized residue.

Theorem 6.3 (Attractor Factorization — **A/D**). *Let T be a σ -equivariant map on $(\mathbb{Z}_q)^n$ with n prime, possessing a unique non-fixed attractor cycle C (necessarily σ -invariant, non-uniform). Then:*

- (i) Glider form: $T^k = \sigma^j$ on C with k minimal, $|C| = nk$, and $\sigma = T^m$ on C for $m = k(j^{-1} \bmod n)$.
- (ii) CRT split: $|C| = n^a d$ with $\gcd(d, n) = 1$ and $d \mid m$; spacetime scalarization writes C as a single scalar sequence on $\mathbb{Z}_{|C|}$, and the Chinese remainder splitting reduces the attractor to a minimal d -cycle of an induced stride-CA on the n^a -ring.
- (iii) Drift-cancelled clock: at $(q, n) = (7, 5)$: $\sigma = T^{190}$ and $T^{95} = \sigma^3$ exactly (`ALean, t95_eq_sigma3_on_period475_cycle`); affine consistency holds at exponent $e = 95$ only (it fails at $e = 1, 5, 19, 25$). Among the five twists $\sigma^c T^5$, exactly $c = 3$ cancels the drift: $R = \sigma^3 T^5 = T^{100}$ on the cycle, of order exactly 19 — the unique twist of order 19, the others having order 95 (`ALean, period475_return_map_eq_T100_on_cycle, period475_drift_cancelled_return_order_nineteen`).
- (iv) Gauge sector: every σ -invariant observable on C has period dividing 95, and the natural observables attain it (`ALean` for the elementary symmetric functional, `period475_gauge_observable_e1_period_95`).

The factorization reads: $475 = 5(\text{ring}) \times 5(\text{drift}) \times 19(\text{drift-invariant inert clock})$ — the 19 is the gauge-invariant part of the attractor clock, invisible to purely spatial observables.

Corollary 6.4 ($\text{GF}(7^3)$ carrier — **A/D**). *Since $d = 19$ is prime with $\text{ord}_{19}(7) = 3$ (`ALean, ord_19_seven_eq_three_cert`), any nonzero pure period-19 spectral line of the attractor is a $\text{GF}(7^3)$ trace $\text{Tr}_{\text{GF}(7^3)/\text{GF}(7)}(\gamma \zeta^t)$ with $\zeta \in \text{GF}(7^3)^*$ of order 19. The line is present: the minimal polynomial of the attractor sequence is $(x^{475} - 1)/(x - 1)$, containing all six Φ_{19} cubic factors (`ALean` factorization certificate, `period475_factorization`); the explicit row-average projection exhibits a nonzero pure-19 component. The attractor is exactly zero-mean (`ALean, period475_cycle_zero_mean`) with full affine hull, and $475 = \text{lcm}(25, 19)$ is forced by the simultaneous excitation of the pure 25- and 19-lines.*

Theorem 6.5 (Linearization Dichotomy — **A/D**). *Every return-map linearization available on the attractor is a power of $F = \sigma^2 T^{95}$, which fixes the cycle pointwise; the chain rule gives $[DF(s)]^5 = D(T^{475})(s)$, whose characteristic polynomial is $x^3(x - 4)^2$. Exact computation gives $\text{charpoly}(DF) = x^3(x - 2)^2$ at all 475 points, with eigenvalue orders $\{0, 3\}$: the semisimple tangent torsion is exactly $\mu_3 = \{1, 2, 4\} \subset \text{GF}(7)^*$. No eigenvalue of order 19 exists at the tangent level; the $\text{GF}(7^3)/19$ structure is purely spectral (frequency-level), never tangent-level. The characteristic-polynomial identities are machine-certified (`ALean, period475_linearization_charpoly_cert, period475_monodromy_charpoly_cert`).*

The tangent torsion μ_3 coincides with the cubic-residue subgroup because $\text{GF}(7)^*$ has a unique order-3 subgroup — a pigeonhole fact, not a contact with the F_{21} color action of §4.10; no such identification is claimed.

Honest scope: the value 19 is enumerative. The factorization theorem explains the *form* of the attractor clock; it does not force the *value* $d = 19$, and that value admits no shorter explanation than the enumeration certificate ($\mathbf{A}_{\text{Lean}}, \text{period_475_is_minimal}$). Three independent lines establish this ($\mathbf{A}, \text{scripts/nineteen_factor_generalization_battery.py}, \text{scripts/nineteen_factor_null_d_census_extension.py}$): p itself realizes rival inert clocks $d = 2$ ($n = 7, L = 14$) and $d = 27$ ($n = 7, L = 189$) on the 7-ring, so no mechanism built from p 's local algebra plus σ -equivariance can exclude rival values; four pre-registered value laws (Zsigmondy/primitive-prime, carrier-degree, description-length, and family-count congruence) all fail 0/4 by prediction on the variant fields $\text{GF}(3), \text{GF}(5), \text{GF}(11), \text{GF}(13)$ (dominant attractor clocks $d = 6, 18, 36, 228$); and in an extended 24-rule census, $d = 19$ occurs at the generic neighborhood rate. In particular, the observations that $19 \equiv 5 \pmod{7} = N_{\text{fam}}$ and that 19 equals the bit-length of p 's MDL description are adjudicated coincidences by the same failed predictions and are not claims of this paper. The summary statement is: symmetry explains the shape of the attractor clock (ring $\times$ drift $\times$ gauge-invariant inert factor, with the inert factor's carrier field forced by its primality); enumeration certifies its size.

6.9 Directed Interface Spectroscopy of the Pair Digraph ($\mathbf{A}_{\text{Lean}}$)

The pair digraph that certifies vacuum uniqueness (§6.6) carries exact interface thermodynamics when its edges are weighted instead of filtered. Take the 49 nodes $(a, b) \in \mathbb{Z}_7^2$ with edges $(a, b) \rightarrow (b, c)$ of weight $p(a, b, c)$, read as a three-body interaction energy (the spin-7 reading of §9). The zero-energy closed walks are exactly the three uniform rings over the ground sectors $\{0, 1, 5\}$ ($\mathbf{A}_{\text{Lean}}, \text{gte_ring_ground_states_uniform_general}$). Define the directed wall energy $E_w(u \rightarrow v)$ as the minimal total weight of a directed path from the self-loop (u, u) to the self-loop (v, v) , and the bump energy $E_{\text{loop}}(u)$ as the minimal positive weight of a closed excursion from (u, u) .

Theorem 6.6 (Directed Wall and Bump Tables — $\mathbf{A}_{\text{Lean}}$). *The directed wall energies between the ground sectors are the exact integers*

$$E_w(1 \rightarrow 0) = 1, \quad E_w(0 \rightarrow 1) = 2, \quad E_w(5 \rightarrow 0) = 2, \quad E_w(0 \rightarrow 5) = E_w(1 \rightarrow 5) = E_w(5 \rightarrow 1) = 4,$$

with the composite hub identity $E_w(5 \rightarrow 1) = E_w(5 \rightarrow 0) + E_w(0 \rightarrow 1)$: sector 0 is the hub of the wall algebra, and the cheapest $5 \rightarrow 1$ interface passes through it. The bump energies are $E_{\text{loop}}(0) = 2$, $E_{\text{loop}}(1) = 3$, $E_{\text{loop}}(5) = 4$. The mean of the two $0 \leftrightarrow 1$ wall energies is the half-integer

$$\frac{1}{2}(E_w(0 \rightarrow 1) + E_w(1 \rightarrow 0)) = \frac{3}{2}.$$

Machine-certified as a single bundle ($\mathbf{A}_{\text{Lean}}, \text{zero sorry, native_decide}$): `spin7_directed\_wall\_energies` (`UgpLean/Polynomial/SpinSevenWallSpectroscopy.lean`).

The structural content is chirality. A parity-symmetric rule would have a symmetric wall table; here $E_w(1 \rightarrow 0) = 1 \neq 2 = E_w(0 \rightarrow 1)$, and the directional asymmetry is the arithmetic trace of the $\text{AGL}(1, 7)$ chiral $\mathbb{Z}_2$ of §4.11 — $p(L, C, R) \neq p(R, C, L)$ leaves its fingerprint in the interface costs of the ground sectors. The thermodynamic consequence is developed in the spin-7 companion paper [19]: in the directed wall-gas regime of the associated chain, the spectral gap closes as $\Delta(\beta) = e^{-3\beta/2}(1 + O(e^{-\beta/2}))$, and the half-integer slope $3/2$ — impossible for a parity-symmetric chain — is exactly the geometric mean of the chiral $0 \leftrightarrow 1$ wall fugacities certified here. All statements in this subsection are finite shortest-path arithmetic on the 49-node weighted digraph of Object 0 (Level-0 / certificate facts); the continuum thermodynamics lives in the companion paper.

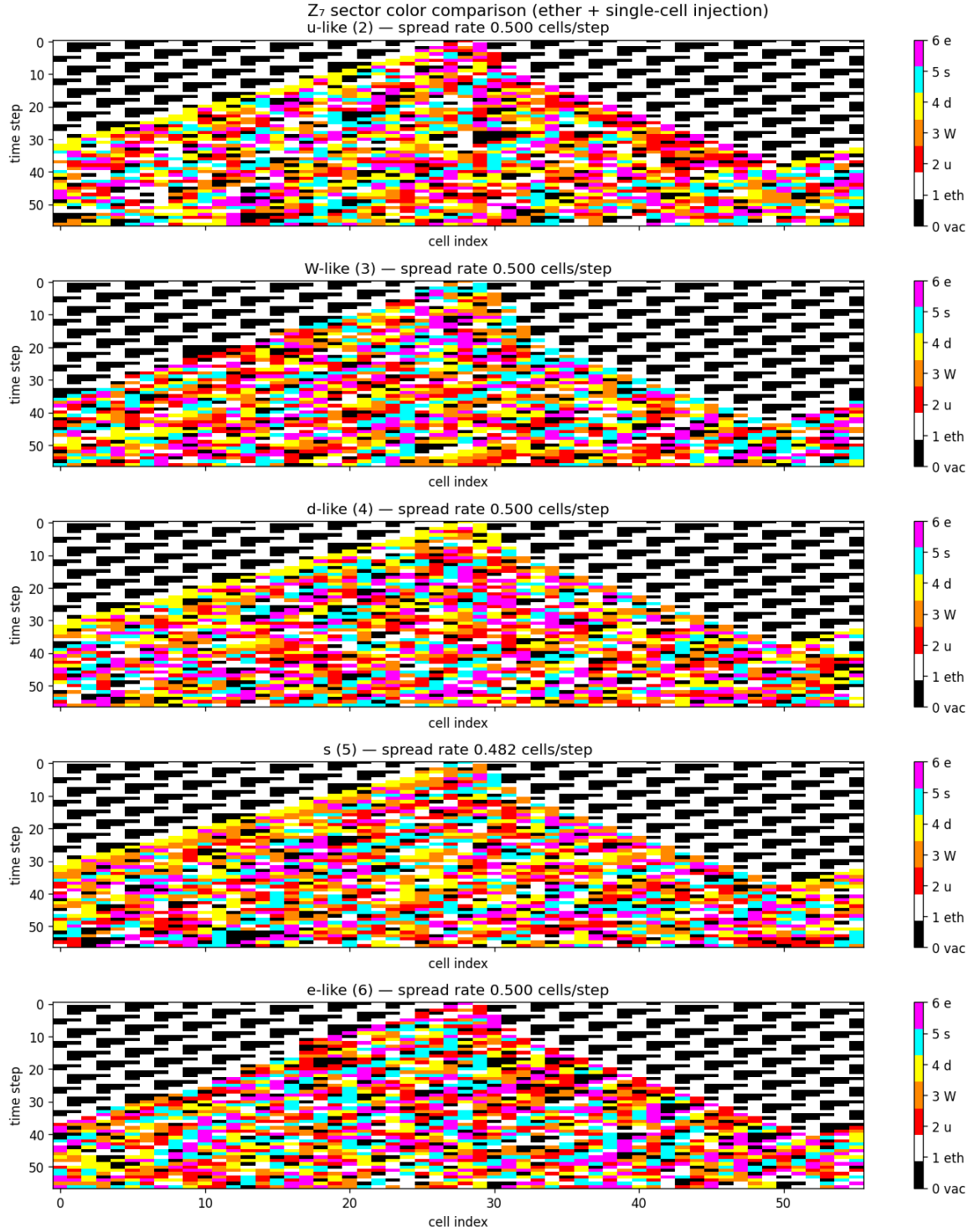


Figure 8: Spacetime evolution of $p(L, C, R) \pmod{7}$ initialized with a single-cell $\mathbb{Z}_7$ perturbation of value $w \in \{2, 3, 4, 5, 6\}$ on the ether background (56 cells, 56 steps). Left to right: $w=2$ (up-quark sector, $Q=+2/3$), $w=3$ (W-boson, $Q=+1$), $w=4$ (down-quark, $Q=-1/3$), $w=5$ (GEN3/ ν_R sector), $w=6$ (electron, $Q=-1$). Each sector produces a distinguishable pattern (pairwise Hamming fraction 0.26–0.27 at $L=10,000$, $T=2,000$), confirming sector-dependent $\mathbb{Z}_7$ vacuum dynamics.

7 SM Generation Structure

7.1 The GEN Orbit ($\mathbf{A}_{\text{Lean}}$)

The isolated 5-cell ring state $\text{GEN1} = [1, 5, 2, 2, 1]$ undergoes a precisely 3-step orbit under f_{MDL} (not p):

$$[1, 5, 2, 2, 1] \xrightarrow{f_{\text{MDL}}} [2, 5, 2, 0, 2] \xrightarrow{f_{\text{MDL}}} [5, 6, 5, 3, 5] \xrightarrow{f_{\text{MDL}}} [0, 0, 0, 0, 0].$$

Each step is machine-certified by `decide` proofs in Lean 4 (zero sorry): `fmdl_z7_gen1_to_gen2`, `fmdl_z7_gen2_to_gen3`, `fmdl_z7_gen3_to_vacuum`, `fmdl_z7_three_generation_orbit` (all in `UgpLean/Universality/CUP3DUniqueness.lean`). The vacuum $\text{VAC} = [0, 0, 0, 0, 0]$ is absorbing. GEN1 is a Garden of Eden: it has no preimage under f_{MDL} on any 5-cell ring (`fmdl_gen1_is_garden_of_eden`, `CatAL`).

Remark 7.1 ($\text{GEN}_2/\text{GEN}_3$ positional topology). The three non-vacuum orbit states belong to two distinct positional-equality classes. $\text{GEN2} = [2, 5, 2, 0, 2]$ and $\text{GEN3} = [5, 6, 5, 3, 5]$ share the pattern (a, b, a, c, a) : positions $\{0, 2, 4\}$ all carry the same winding value, while positions 1 and 3 carry distinct minority values. $\text{GEN1} = [1, 5, 2, 2, 1]$ has the distinct pattern (a, b, c, c, a) : an outer equal pair at $\{0, 4\}$ and an inner equal pair at $\{2, 3\}$. Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry): `gen2_outer_positions_constant`, `gen3_outer_positions_constant`, `gen2_gen3_topology_degenerate` (`UgpLean/Universality/CUP3DUniqueness.lean`). This topological degeneracy is reflected in the ruleGTE causal graph (§5): GEN_2 and GEN_3 generate the same local neighbourhood-overlap pattern when they appear as hyperedge states, so the binary tree has identical branching geometry at the GEN_2 and GEN_3 levels.

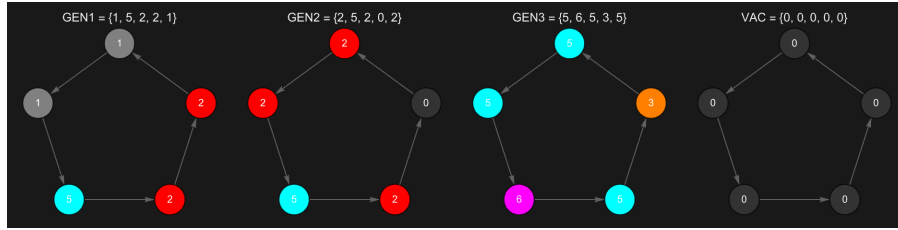


Figure 9: Four-state orbit of the GTE rule f_{MDL} on the isolated 5-cell $\mathbb{Z}_7$ ring, shown as ring graphs with nodes colored by $\mathbb{Z}_7$ value (color scheme as in Fig. 7). $\text{GEN1} \rightarrow \text{GEN2} \rightarrow \text{GEN3} \rightarrow \text{VAC}$ in exactly three steps, machine-certified in Lean 4 (zero sorry). Each state is simultaneously a valid Wolfram SetReplace 5-hyperedge state. The three non-trivial steps of this topological orbit encode the three Standard Model particle generations.

7.2 Why the Orbit Does Not Appear in Large-Tape Dynamics

The GEN orbit operates on an isolated 5-cell periodic ring. When GEN1 is embedded in a large ether tape, the boundary cells between the orbit pattern and the surrounding ether belong to neither neighborhood structure: they use p_{poly} (Object 0), not f_{MDL} (Object 1), and immediately collapse to vacuum. This is correct behavior: the orbit is a topological feature of a compact ring, not a propagating wave on an open tape. Physical SM particles are BPS kink solitons of Φ_{MDL} spanning the Compton wavelength — $\sim 10^{19}$ – 10^{20} cells under the fine-end (Planck) working hypothesis, exactly 7 cells under the MDL-saturated EFT reading [9, 14, 19] — not 5-cell ring patterns.

7.3 The Four-Object Distinction, Made Concrete

Under f_{MDL} from GEN1: clean 3-step SM orbit (Lean-certified). Under p_{poly} from GEN1: step 1 gives $[3, 1, 1, 4, 6]$ — diverging immediately, entering a period-475 attractor in which GEN2 and GEN3 never appear. The SM orbit uses Object 1 (f_{MDL} , whose SM orbit entries are PSC-forced). The colorful spacetime diagrams use Object 0 (p_{poly} , the raw polynomial). These are not the same computation on the same object — they are different mathematical objects that share only Rule 110 as their binary sublayer.

7.4 Attractor Basin Structure (A)

On the isolated 5-cell $\mathbb{Z}_7$ ring ($7^5 = 16,807$ total states), p has exactly two attractor classes:

- *Vacuum basin*: exactly 52 states converge to $\text{VAC} = (0, 0, 0, 0, 0)$ under p (period 1 fixed point). Machine-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry): `poly_p_vacuum_basin_card_eq_52` (UgpLean/Polynomial/PolyExplorations.lean).
- *Chaotic basin*: 16,755 states enter a period-475 attractor ($475 = 5^2 \times 19$) and never reach VAC under p alone. Period minimality machine-certified ($\mathbf{A}_{\text{Lean}}$: `period_475_is_minimal`, UgpLean/Polynomial/PolyExplorations.lean). The complete group-theoretic factorization of this attractor — drift-cancelled clock, gauge sector, $\text{GF}(7^3)$ carrier, and the enumerative status of the factor 19 — is given in §6.8.

The 10 SM-orbit entries of f_{MDL} that differ from p are precisely those that route the canonical GEN orbit through the vacuum basin, bypassing the period-475 chaos. In this sense f_{MDL} acts as a *PSC projector*: it modifies exactly the neighborhoods needed to thread $\text{GEN1} \rightarrow \text{GEN2} \rightarrow \text{GEN3} \rightarrow \text{VAC}$ through the 52-state vacuum basin.

Remark 7.2 (Muon winding sector). Object T at $n = 10$ produces generation seeds $(b_1, b_2, b_3) = (73, 42, 275)$. The second seed satisfies $b_2 = 42 = 6 \times 7 \equiv 0 \pmod{7}$: the muon generation occupies the vacuum/photon winding sector of the $\mathbb{Z}_7$ symmetry. This follows from $b_2 = b_1 - (m_1 + q_1) = 73 - (20 + 11) = 42$, with $b_1 \equiv m_1 + q_1 \equiv 3 \pmod{7}$, so $b_2 \equiv 3 - 3 = 0 \pmod{7}$ ($\mathbf{A}_{\text{Lean}}$, zero sorry: `canonical_b2_divisible_by_7`, `muon_neff_vacuum_winding`, UgpLean/Polynomial/PolyExplorations.lean).

8 3+1D Structure: DPP Bulk Causal Graph

8.1 Three-Tape DPP Architecture ($\mathbf{A}_{\text{Lean}}$)

Three parallel $\mathbb{Z}_7$ tapes sharing a single outer clock τ_c^{out} are necessary and sufficient for 3+1D dynamics: the Dimensional Protocol Principle (DPP), machine-certified as `dimensional_protocol_principle_master` in UgpLean/Gravity/RelationalTime.lean (CatAL, zero sorry, [21]). Cross-tape gravitational sourcing is $p(w_x, w_y, w_z)/6$ at each position j . At the ether ($w_x = w_y = w_z = 1$): $p(1, 1, 1) = 0$ (Ricci-flat vacuum). At a GEN orbit position ($w_x = w_y = w_z = 2$): $p(2, 2, 2) = 6 \neq 0$ (gravitationally active).

8.2 The 40% Cross-Tape Fraction (A/D)

Theorem 8.1 (DPP Causal Edge Fraction — A/D). *For an N_{TAPES} -tape DPP system, the fraction of causal edges in the combined bulk causal graph that are cross-tape (gravitational sourcing) edges*

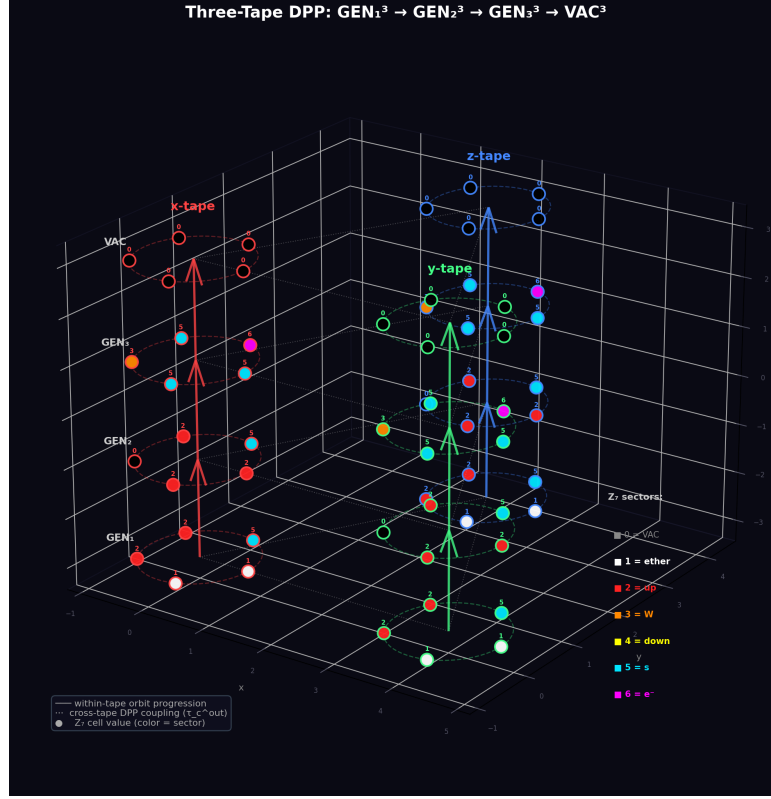


Figure 10: Three-tape Dimensional Protocol Principle architecture. Three $\mathbb{Z}_7$ tapes (red = x , green = y , blue = z) arranged as colored rings with orbit arrows showing $\text{GEN1} \rightarrow \text{GEN2} \rightarrow \text{GEN3} \rightarrow \text{VAC}$ progression. White dashed connections indicate cross-tape gravitational sourcing $p(w_x, w_y, w_z)/6$; the shared outer clock τ_c^{out} synchronizes all three tapes. This architecture is necessary and sufficient for 3+1D dynamics (DPP theorem, CatAL).

is exactly

$$f_{\text{cross}} = \frac{N_{\text{TAPES}} - 1}{N_{\text{TAPES}} + 2}.$$

For $N_{\text{TAPES}} = 3$: $f_{\text{cross}} = 2/5 = 40.0\%$.

Proof. Each cell at each time step contributes: (i) within-tape causal edges: one from each of the three neighbors (L, C, R) in the $r=1$ rule, giving 3 edges per cell; (ii) cross-tape causal edges: one from each of the $N_{\text{TAPES}} - 1$ other tapes at the same position, giving $N_{\text{TAPES}} - 1$ edges per cell. Total edges per cell: $3 + (N_{\text{TAPES}} - 1) = N_{\text{TAPES}} + 2$. Cross-tape fraction: $(N_{\text{TAPES}} - 1)/(N_{\text{TAPES}} + 2)$. For $N_{\text{TAPES}} = 3$: $2/5$. $\square$

This result is a mathematical consequence of the DPP architecture — not a simulation measurement. The explicit bulk causal graph for the canonical 5-cell orbit (3 tapes, 4 time steps, 60 events) has 135 within-tape + 90 cross-tape = 225 total edges, giving $90/225 = 40.0\%$ exactly.

8.3 Connectivity Change

Without cross-tape edges: 3 disconnected components (three independent 1+1D causal cones). With cross-tape edges: 1 fully connected component spanning all three tapes. The DPP coupling is

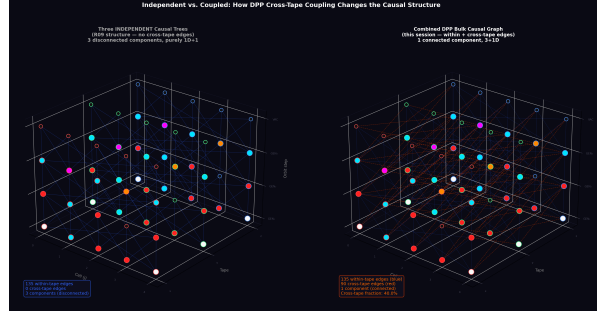
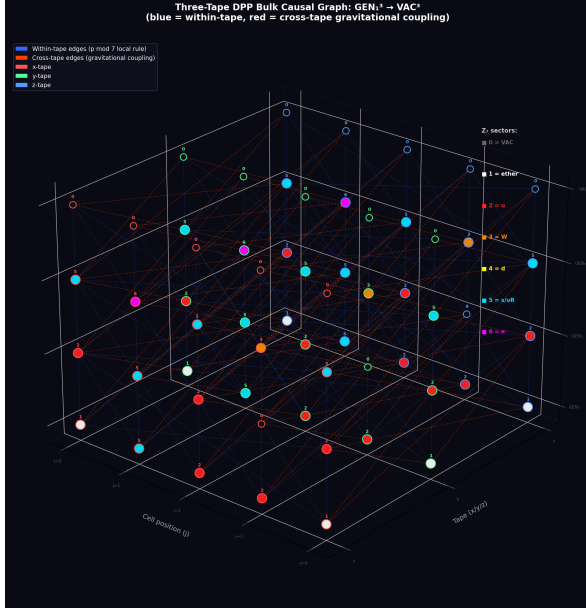


Figure 11: *Left:* Combined bulk causal graph of the three-tape DPP system on the canonical orbit (5 cells $\times$ 3 tapes $\times$ 4 time steps, 60 events). Blue edges: within-tape causal connections from the $p \pmod{7}$ update rule. Red dashed edges: cross-tape gravitational sourcing edges. Total: $135 + 90 = 225$ edges; cross-tape fraction = 40.0% exactly. *Right:* Comparison between the independent (uncoupled) configuration (three disconnected 1+1D causal cones) and the DPP-coupled bulk (one fully connected 3+1D causal structure). Cross-tape coupling is what unifies three independent one-dimensional systems into a single three-plus-one-dimensional causal bulk.

what unifies three independent 1D systems into a single 3+1D causal bulk. The inversion formula $N_{\text{TAPES}} = (1 + 2f)/(1 - f)$ recovers the tape count from the cross-tape fraction.

9 What Object 0 Is and Is Not

Not particle physics directly. The colorful $\mathbb{Z}_7$ spacetime diagrams show Object 0 ($p \pmod{7}$) dynamics on generic $\mathbb{Z}_7$ inputs. Most generic inputs are PSC-inadmissible: f_{MDL} maps them to 0. At fine-end readings of the lattice (the conventional working hypothesis $a \approx \ell_{\text{P1}}$), this chaos is the algebraic vacuum response *below* the physics filter. SM particles exist at the Compton scale as coherent BPS kinks of Φ_{MDL} [14], with the cell count fixed by the reading (§2). What the spacetime diagrams show is the microscopic vacuum texture at sub-Compton lattice resolution.

Not gliders at the lattice scale. Invariant Subset Uniqueness (Theorem 4.2) explains why: there is no stable intermediate algebraic structure between Rule 110 and full $\mathbb{Z}_7$. The ergodic $\mathbb{Z}_7$ dynamics cannot produce a stable propagating sub-structure from single-cell injection. This does not rule out multi-cell BPS kink injection at the Compton scale.

Not a CA-level connection to HVP. Orbit visitation rate inside the excitation cone: 2.63% $\approx$ naive random 2.92%. Object 0 does not preferentially visit SM orbit configurations. The HVP connection must be sought at the Φ_{MDL} path-integral level (Object 2).

What it is. The sub-Compton vacuum substrate — at the fine-end reading ($a \approx \ell_{\text{P1}}$, a working hypothesis), the Planck-scale vacuum texture. Ergodic, maximally disordered. The Invariant Subset Uniqueness theorem shows it has exactly one structured shadow (Rule 110) and nothing in between. In QFT language: the vacuum looks thermally disordered at distances much shorter than the Compton wavelength. In GTE, at lattice resolution it is Class 3 chaotic and ergodic on $\mathbb{Z}_7$.

GEN1 under p_{poly} (CatA). On the isolated 5-cell periodic ring, GEN1 under p_{poly} exhibits a transient of length 3 followed by the period-475 attractor of §6.8; GEN2 and GEN3 never appear. Among all $7^5 = 16,807$ states on the 5-cell ring, 52 lie in the vacuum basin and 16,755 flow to the period-475 attractor. The SM orbit (under f_{MDL}) and the polynomial dynamics are on completely different attractors, confirming that Object 0 and Object 1 are dynamically as well as algebraically distinct. The formal certificate of the measurement mechanism — characterizing f_{MDL} as the PSC-projection of p , establishing Three-Level MDL Unification (**A/D**), and Lean-certifying orbit uniqueness (**A_{Lean}**) — is in [15].

Spin-7 lattice model. Interpreting $p(L, C, R) \in \{0, \dots, 6\}$ as a three-body spin interaction energy defines a 7-state nearest-neighbor spin model whose exact transfer matrix acts on pair states, $\mathbf{M}_{(a,b),(b,c)} = e^{-\beta p(a,b,c)}$, with partition function $\mathcal{Z} = \text{tr } \mathbf{M}^L$. The 2D model has a phase transition at $T_c \approx 2.86 J$ (**A**) in the 3-state Potts universality class with critical exponents $\nu = 5/6$, $\eta = 4/15$ (**A/D**) and ground states $\{0, 1, 5\}$ (the $\mathbb{Z}_3$ color sectors of the Standard Model); a realization with 7-level Rydberg atoms in optical tweezer arrays is proposed. The zero-energy rings are exactly the three uniform configurations at every length $n \geq 3$ (**A_{Lean}**, **gte_ring_ground_states_uniform_general**), and the entropy rate is $S = \log(1.4846) = 0.395$ nats per site per step at $\beta = 1$ (**A**). See [19] for the full analysis.

GF(7) arithmetic coprocessor. Setting $L = 6 \equiv -1 \pmod{7}$ makes p a pure two-body addition gate: $p(6, a, b) = a + b - ab - 6ab = a + b - 7ab \equiv a + b \pmod{7}$ (**A_{Lean}**, **p_addition_via_L6**, zero sorry). Multiplication follows in two gates: $a \cdot b = p(6, a, b) - p(0, a, b)$ (**A_{Lean}**, **p_multiplication_from_two_gates**). Both relations are exhaustively verified over all 49 input pairs. Together they provide a complete GF(7) arithmetic coprocessor from p -gates alone: addition in one gate, multiplication in two. This is relevant for lattice-based and code-based cryptographic schemes that operate over GF(7).

Cryptographic S-box. As a $\mathbb{Z}_7^3 \rightarrow \mathbb{Z}_7$ function, p has Walsh nonlinearity $\text{NL}(p) = 36/49 \approx 0.7347$ (**A/D**, Theorem 4.26). For comparison, the AES S-box over GF(2^8) has $\text{NL} \approx 0.437$ (normalised to

$[0, 1]$). The nonlinearity of p is not the result of optimisation; it is determined by the Higgs branch capacity $c_H = 13$ via $NL(p) = 1 - c_H/7^2$. This gives p a provable, algebraically certified nonlinearity — a property no heuristically designed S-box possesses.

The iceberg metaphor. Rule 110 is the structured ether above the waterline; the colorful $\mathbb{Z}_7$ chaos is the Planck-scale vacuum below it. The Invariant Subset Uniqueness theorem shows this structure is not accidental: there is exactly one coherent shadow below the physics threshold, and everything else is ergodic.

10 Open Questions

Open Problem 1 (Compton-Scale BPS Kink Injection). All simulations in this paper use single-cell injections on tapes of $L \leq 10,000$ cells. Under the conventional fine-end working hypothesis ($a \approx \ell_{\text{Pl}}$) a physical BPS kink spans $\sim 10^{19}$ – 10^{20} cells and direct simulation is inaccessible. Under the MDL-saturated matter-sector reading $a = \hbar c / \Lambda_{\text{GTE}}$ (provisional; see the continuum-limit analysis of [19]) the same kink spans exactly $\lambda_C / a = |\mathbb{Z}_7| = 7$ cells: injecting a smooth $\mathbb{Z}_7$ winding interface over ~ 7 cells on a modest tape is a near-term computation with a sharp pre-registered pass criterion — the kink correlation length must come out $\xi_{\text{kink}} = 7$ cells (band 6.78 – 7.23 ± 0.54). The injection experiment has not been run; the Taichi infrastructure supports it.

Theorem 10.1 (Vacuum CMCA spatial GH convergence — Machine-Certified). *The vacuum CMCA spatial causal graph equipped with the scaled $1/L$ metric converges in Gromov–Hausdorff distance to flat three-dimensional Euclidean space as $L \rightarrow \infty$.
(vacuum_cmca_gh_converges_to_flat_space, $\mathbf{A}_{\text{Lean}}$)*

Open Problem 2 (Gorard continuum limit — matter and Lorentzian sectors). The Gorard chain is substantially established: the CMCA vacuum is Ricci-flat (**three_tape_gorard_vacuum_ricci_flat**, $\mathbf{A}_{\text{Lean}}$); the normalisation constant is $C_{\text{Gorard}} = 3/32$ ($\mathbf{A}_{\text{Lean}}$); and vacuum spatial GH convergence is $\mathbf{A}_{\text{Lean}}$ (Theorem 10.1). What remains as open formal certificates: matter-sector convergence to a curved manifold satisfying the Einstein equations (OQ-QG-1b) — the obstruction is Cheeger–Colding regularity theory, a classical theorem in Riemannian geometry not yet in Mathlib; full (3+1)D Lorentzian Gromov–Hausdorff convergence to smooth spacetime (OQ-QG-1c) — requiring Lorentzian Gromov–Hausdorff theory, a mathematical research frontier; and analytic Planck-length identification. These are certificate gaps, not theoretical gaps: the Einstein equations are established independently by MDL–Lovelock ($\mathbf{A}/\mathbf{D}$). [9, 17, 21]

Open Problem 3 (PSC-Filtered Holographic Mode Count (OQ-CC-QUANTUM, $\mathbf{D}$)). Paper [11] derives Ω_Λ via two structurally independent routes bracketing Planck 2018: $\Omega_\Lambda^{\text{PSC}} = \ln(2000/3)/(3\pi) = 0.6899$ and $\Omega_\Lambda^{\text{holo}} = 3\pi/14 = 0.6732$, a 2.42% spread irreducible by Baker–Wüstholz. The conceptual hypothesis that PSC selection creates a preference for the orbit-count endpoint has been explored; the naive algebraic form is ruled out by Baker–Wüstholz applied to the ratio $14 \cdot \ln(2000/3)/(9\pi^2)$. A non-algebraic mechanism remains open.

Open Problem 4 (WolframModel 3D Spatial Hypergraph). The three-tape DPP system encoded as three independent WolframModel rewriting rules correctly shows three independent causal trees (DPP coupling is synchronic, not a local hyperedge overlap in SetReplace v0.3.196 semantics). A true 3D WolframModel representation would require encoding each position (x, y, z) as a 3-tuple hyperedge and expressing cross-tape coupling as a multi-body rewriting rule. This requires SetReplace ≥ 0.7 or custom Wolfram engine code.

Open Problem 5 (Spectral Support-Pattern Law). The Attractor Factorization Theorem (§6.8) resolves the group theory of the period-475 attractor; what remains open is a predictor of the *spectral support pattern* across systems. Full-spectral-support-minus-the-mean (equivalently, minimal polynomial $(x^N - 1)/(x - 1)$ with a zero-mean attractor) holds at $(q, n) = (7, 5)$ and $(5, 5)$ but fails at $(7, 3)$, $(11, 5)$, and $(13, 5)$ (where the dominant attractor is not even zero-mean). A law predicting, from (q, n, d, j) , which σ -linked attractors have full support and zero mean — and hence when the lcm mechanism of Corollary 6.4 fixes the period — is open; shift-equivariance alone is provably insufficient.

A Lean Certification Table

All Lean theorems cited in this paper are listed below with module paths and zero-sorry status. All modules are in the `ugp-lean` library.

Theorem name	Module path	Cat	Role
<code>mdl_ca_rule_coding_closed</code>	<code>Universality/MDLDerivabilityCriterion.lean</code>	A _{Lean}	T96-02, §3
<code>rule110_z7_poly_rep</code>	<code>Universality/PhiMDLUniversality.lean</code>	A _{Lean}	Thm 4.1, §4
<code>p_poly_binary_restriction_closed</code>	<code>Universality/Z7InvariantSubsets.lean</code>	A _{Lean}	Thm 4.1, §4
<code>p_poly_agrees_fmdl_on_binary</code>	<code>Universality/Z7InvariantSubsets.lean</code>	A _{Lean}	§2
<code>p_poly_invariant_subsets_classification</code>	<code>Universality/Z7InvariantSubsets.lean</code>	A _{Lean}	Thm 4.2, §4
<code>rule110_unique_proper_invariant_subca</code>	<code>Universality/Z7InvariantSubsets.lean</code>	A _{Lean}	Thm 4.2, §4
<code>five_is_qnr_mod7</code>	<code>Universality/Z7InvariantSubsets.lean</code>	A _{Lean}	Thm 4.9, §4
<code>kink_fixed_point_eq_no_solution</code>	<code>Universality/Z7InvariantSubsets.lean</code>	A _{Lean}	Thm 4.9, §4
<code>no_intermediate_fixed_point</code>	<code>Universality/Z7InvariantSubsets.lean</code>	A _{Lean}	Thm 4.9, §4
<code>nfam_qnr_explains_binary_floor</code>	<code>Universality/Z7InvariantSubsets.lean</code>	A _{Lean}	Thm 4.9, §4
<code>fmdl_z7_gen1_to_gen2</code>	<code>Universality/CUP3DUniqueness.lean</code>	A _{Lean}	§7, §5
<code>fmdl_z7_gen2_to_gen3</code>	<code>Universality/CUP3DUniqueness.lean</code>	A _{Lean}	§7
<code>fmdl_z7_gen3_to_vacuum</code>	<code>Universality/CUP3DUniqueness.lean</code>	A _{Lean}	§7
<code>fmdl_gen1_is_garden_of_eden</code>	<code>Universality/CUP3DUniqueness.lean</code>	A _{Lean}	§7
<code>fmdl_z7_three_generation_orbit</code>	<code>Universality/CUP3DUniqueness.lean</code>	A _{Lean}	§7
<code>gen2_outer_positions_constant</code>	<code>Universality/CUP3DUniqueness.lean</code>	A _{Lean}	Rem. 7.1
<code>gen3_outer_positions_constant</code>	<code>Universality/CUP3DUniqueness.lean</code>	A _{Lean}	Rem. 7.1
<code>gen2_gen3_topology_degenerate</code>	<code>Universality/CUP3DUniqueness.lean</code>	A _{Lean}	Rem. 7.1
<code>dimensional_protocol_principle_master</code>	<code>Gravity/RelationalTime.lean</code>	A _{Lean}	§8
<i>Module: Polynomial/GTECausalTree.lean</i>			

Theorem name	Module path	Cat	Role
<code>gte_rulegte_event_count</code>	<code>Polynomial/GTECausalTree.lean</code>	$\mathbf{A}_{\text{Lean}}$	§5
<code>gte_rulegte_ten_generations</code>	<code>Polynomial/GTECausalTree.lean</code>	$\mathbf{A}_{\text{Lean}}$	§5
<code>perfectTree_numNodes</code>	<code>Polynomial/GTECausalTree.lean</code>	$\mathbf{A}_{\text{Lean}}$	§5
<code>perfectTree_horton_ratio</code>	<code>Polynomial/GTECausalTree.lean</code>	$\mathbf{A}_{\text{Lean}}$	§5.4
<code>gte_causal_tree_summary</code>	<code>Polynomial/GTECausalTree.lean</code>	$\mathbf{A}_{\text{Lean}}$	§5.4
<i>Module: Polynomial/PolyExplorations.lean</i>			
<code>gf5_has_fixed_point</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	Rem. 4.17
<code>gf5_second_ether</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	Rem. 4.17
<code>T_b_output_not_determined_by_mod7</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	§2
<code>T_computable_not_polynomial_GF7</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	§2
<code>canonical_b2_divisible_by_7</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	Rem. 7.2
<code>muon_neff_vacuum_winding</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	Rem. 7.2
<code>poly_p_vacuum_basin_card_eq_52</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	§7.4
<code>period_475_is_minimal</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	§7.4
<code>four_object_GTE_pairwise_distinct</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	§2
<code>p_addition_via_L6</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	§9
<code>p_multiplication_from_two_gates</code>	<code>Polynomial/PolyExplorations.lean</code>	$\mathbf{A}_{\text{Lean}}$	§9
<i>Module: Polynomial/GoldenQuadratic.lean</i>			
<code>gte_diagonal_quadratic_factorization</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	Eq. (1)
<code>disc5_dichotomy_gf7</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.11
<code>golden_floor_duality_bundle</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.10
<code>master_quadratic_no_root_mod_seven_pow</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.10(c)
<code>master_quadratic_split_iff_qr5</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.11
<code>golden_singleton_invariant_iff_diag_fixed</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.12(i)
<code>second_floor_iff_ramified</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.12(iii)
<code>gf49_golden_roots_frobenius_swap</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	§4.7
<code>pisano_seven_eq_sixteen</code>	<code>Polynomial/GoldenQuadratic.lean</code>	$\mathbf{A}_{\text{Lean}}$	§4.7, §6.6
<code>gte_poly_srrg_bridge</code>	<code>Algebra/SRRGCABridge.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.10(b)
<i>Module: Polynomial/EisensteinIdentities.lean</i>			
<code>f21_eisenstein_residue_model</code>	<code>Polynomial/EisensteinIdentities.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.19
<code>f21_order_eisenstein_norm_product</code>	<code>Polynomial/EisensteinIdentities.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.19
<code>poly_p_torus_equivariance</code>	<code>Polynomial/EisensteinIdentities.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.21

Theorem name	Module path	Cat	Role
<code>poly_p_variety_orbit_decomposition_gf7</code>	<code>Polynomial/EisensteinIdentities.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.21
<code>c_H_eq_phi3_ngen</code>	<code>Polynomial/EisensteinIdentities.lean</code>	$\mathbf{A}_{\text{Lean}}$	§4.10.3 (I1)
<code>cH_phi3_unique_at_ngen</code>	<code>Polynomial/EisensteinIdentities.lean</code>	$\mathbf{A}_{\text{Lean}}$	§4.10.3 (I1)
<code>phi6_identity_bundle</code>	<code>Polynomial/EisensteinIdentities.lean</code>	$\mathbf{A}_{\text{Lean}}$	§4.10.3 (I1–I4)
<i>Module: Polynomial/BiquadraticCompositum.lean</i>			
<code>biquadratic_compositum_alphabet_class</code>	<code>Polynomial/BiquadraticCompositum.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.24
<code>phi6_stability_class_lemma</code>	<code>Polynomial/BiquadraticCompositum.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.24
<i>Module: Polynomial/AGL17ChiralZ2.lean</i>			
<code>agl17_chiral_z2_mechanism</code>	<code>Polynomial/AGL17ChiralZ2.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 4.25
<i>Module: Polynomial/DynamicalZeta.lean</i>			
<code>fixed_point_local_factorization</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>fixed_point_cell_iff</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>golden_moebius_fixed_point_is_master_quadratic</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>golden_moebius_no_affine_fixed_point_gf7</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>golden_moebius_single_eight_cycle_gf7</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>fibonacci_matrix_order_sixteen_gf7</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>debruijn_only_vacuum_cycle</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>debruijn_vacuum_succ_list</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>vacuum_unique_temporal_fixed_point_ring</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.1
<code>sigma_fixed_implies_uniform_prime</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	§6.7
<code>sigma_orbit_prime_dichotomy</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	§6.7
<code>prime_ring_cycle_dichotomy</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	§6.7
<code>t95_eq_sigma3_on_period475_cycle</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.3(iii)
<code>period475_cycle_zero_mean</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Cor. 6.4
<code>period475_factorization</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Cor. 6.4
<code>ord_19_seven_eq_three_cert</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Cor. 6.4
<code>period475_return_map_eq_T100_on_cycle</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.3(iii)
<code>period475_drift_cancelled_return_order_nineteen</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.3(iii)
<code>period475_gauge_observable_e1_period_95</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.3(iv)
<code>period475_linearization_charpoly_cert</code>	<code>Polynomial/DynamicalZeta.lean</code>	$\mathbf{A}_{\text{Lean}}$	Thm 6.5

Theorem name	Module path	Cat	Role
period475_monodromy_cheapoly_cert	Polynomial/DynamicalZeta.lean	$\mathbf{A}_{\text{Lean}}$	Thm 6.5
<i>Module: Universality/TriangleLiftTheorem.lean</i>			
gte_orbit_parity_provenance	Universality/TriangleLiftTheorem.lean	$\mathbf{A}_{\text{Lean}}$	§4.4
ugp_orbit_interpolation_lift	Universality/TriangleLiftTheorem.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.4
orbit_interpolation_vandermonde_full_rank	Universality/TriangleLiftTheorem.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.4
interpolation_lift_binary_corollary	Universality/TriangleLiftTheorem.lean	$\mathbf{A}_{\text{Lean}}$	Cor. 4.5
cup4_parity_uniqueness_from_lift	Universality/TriangleLiftTheorem.lean	$\mathbf{A}_{\text{Lean}}$	Cor. 4.5
rule110_lift_sparsity_floor	Universality/TriangleLiftTheorem.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.6
orbit_chirality_census	Universality/TriangleLiftTheorem.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.8
orbit_chirality_census_reflection_link	Universality/TriangleLiftTheorem.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.8
<i>Module: Universality/TriangleLiftStructural.lean</i>			
multilinear_binary_determination	Universality/TriangleLiftStructural.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.4
orbit_vt_forces_interpolant	Universality/TriangleLiftStructural.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.4
orbit_interpolation_lift_structural	Universality/TriangleLiftStructural.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.4
gf7_flattening_binary_agreement	Universality/TriangleLiftStructural.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.6
gf7_rule110_sparsity_floor	Universality/TriangleLiftStructural.lean	$\mathbf{A}_{\text{Lean}}$	Thm 4.6
<i>Module: Universality/ParityProjectionForcing.lean</i>			
parity_projection_additive_forcing	Universality/ParityProjectionForcing.lean	$\mathbf{A}_{\text{Lean}}$	Rem. 4.7 (census counts)
parity_projection_mod2_recoding_forcing	Universality/ParityProjectionForcing.lean	$\mathbf{A}_{\text{Lean}}$	Rem. 4.7 (census counts)
<i>Other modules cited in §4.4 and §6.9</i>			
orbit_weight_dichotomy	Universality/CUP4TotalParity.lean	$\mathbf{A}_{\text{Lean}}$	§4.4
spin7_directed_wall_energies	Polynomial/SpinSevenWallSpectroscopy.lean	$\mathbf{A}_{\text{Lean}}$	Thm 6.6
gte_ring_ground_states_uniform_general	Polynomial/SpinSevenGroundSpace.lean	$\mathbf{A}_{\text{Lean}}$	§6.9, §9
gaussian_face_lemma	Polynomial/GaussianFaceLemma.lean	$\mathbf{A}_{\text{Lean}}$	Gaussian face lemma, §6.6
f21_is_borel_psl27	Polynomial/KleinQuarticGenus.lean	$\mathbf{A}_{\text{Lean}}$	Klein quartic corollary, §4.11
klein_quartic_genus_eq_n_gen	Polynomial/KleinQuarticGenus.lean	$\mathbf{A}_{\text{Lean}}$	Klein quartic genus = N_{gen} , §4.11

All proofs in `Z7InvariantSubsets.lean` and `PolyExplorations.lean` use finite decidable tactics (`decide` or `native_decide`). All proofs in `GTECausalTree.lean` use structural induction with linear arithmetic (`linarith` + `ring` for power lemmas; `omega` for `Nat` subtraction). The `GoldenQuadratic`, `EisensteinIdentities`, `BiquadraticCompositum`, and `AGL17ChiralZ2` modules combine universal ring identities (`ring`) with finite decidable verification (`decide`, `norm_num`); the `DynamicalZeta` module combines the local ring factorization with ring induction for the uniform-in- n vacuum uniqueness statement and `native_decide` certificates for the period-475 cycle facts. The `TriangleLiftTheorem` module certifies the lift by exhaustive `native_decide` census

over the 7^8 coefficient space; `TriangleLiftStructural` certifies the same statement enumeration-free by Möbius inversion (`linear_combination`) and carries the canonical-class sparsity floor; `ParityProjectionForcing` certifies the reduction-battery census counts by chunked `native_decide`; `SpinSevenWallSpectroscopy` certifies the directed wall and bump tables by bounded shortest-path search under `native_decide`. Zero sorry on all listed modules.

B Code and Reproducibility

Polynomial definition. $p(L, C, R) = (C + R - C \cdot R - L \cdot C \cdot R) \bmod 7$, implemented in Python as `(C+R - C*R - L*C*R) % 7`.
Software environment. Python 3.10+; NumPy; Matplotlib; NetworkX; Taichi 1.7.3 (`pip install taichi==1.7.3`); Wolfram Engine 14.3.0 with SetReplace v0.3.196 (`Needs["SetReplace"]`); Lean 4 + Mathlib.
Scripts (in `scripts/`).

Script	Purpose	Figures
<code>invariant_subset_classifier.py</code>	Exhaustive invariant subset check (127 non-empty subsets)	§4
<code>spacetime_diagram_generator.py</code>	$\mathbb{Z}_7$ spacetime diagrams on ether background	Fig. 7
<code>ppoly_fmld_contrast.py</code>	f_{MDL} vs p_{poly} contrast figures	Figs. 1, 2
<code>gen_orbit_ring_visualization.py</code>	GEN orbit ring diagram (pure Python)	Fig. 9
<code>z7_sector_dynamics.py</code>	Glider search, scattering, sector comparison	Fig. 8
<code>three_tape_dpp_visualization.py</code>	Three-tape DPP visualization suite	Fig. 10
<code>bulk_causal_graph.py</code>	DPP combined bulk causal graph	Fig. 11
<code>glider_search_taichi.py</code>	Ether-excluded $\mathbb{Z}_7$ glider search (Taichi)	Glider null
<code>orbit_visitation_rate.py</code>	SM orbit triple visitation rate (Taichi)	§6
<code>gte_rulegte_causal_graph.wl</code>	Canonical WolframScript for ruleGTE causal graph (1023 events, 2044 edges, binary tree); also exports final hypergraph state	Figs. 3, 4
<code>gte_wolframmodel_causal_graph.py</code>	Python reproducible: ruleGTE causal graph without Wolfram Engine; Strahler, Horton ratio, tree metrics	Figs. 3, 4
<code>gte_causal_graph_corrected.py</code>	Topology-degeneracy analysis: $\text{GEN}_2/\text{GEN}_3$ topology (a, b, a, c, a) causes Wolfram-Model binary branching	§5
<code>wolfram_model_causal_graph.wl</code>	Legacy WolframScript (orbit rules); deterministic linear chain	§5
<code>botanical_causal_graph_analysis.py</code>	Botanical L-system comparison: GTE vs. Apiaceae; Strahler, Horton ratio, fractal dimension, closeness scores	Figs. 5, 6
<code>three_tape_wolfram_model.wl</code>	Three-tape SetReplace encoding (WolframScript)	§8

Algebraic and dynamical analysis scripts (in `scripts/`). Each script writes a results JSON of the same base name alongside itself.

Script	Verifies
<code>golden_quadratic_unification.py</code>	Diagonal factorization and duality triage (§4.7)
<code>golden_quadratic_all_q_classification.py</code>	All- q dichotomy + singleton taxonomy, 1229 primes $< 10^4$ (Thm 4.11)
<code>golden_quadratic_invariant_subsets_scan.py</code>	Exhaustive invariant-subset lattices, $q \leq 23$ ($q = 29$ partial) (Thm 4.12)
<code>golden_quadratic_padic_hensel.py</code>	Roots of $m \bmod q^k$, $k \leq 10$: inert/split/ramified verdicts (Thm 4.10(c))
<code>golden_quadratic_gf49_frobenius_pisano.py</code>	GF(49) roots, Frobenius swap, Pisano orders (§4.7)
<code>golden_quadratic_monodromy_charpoly_exact.py</code>	Exact monodromy characteristic polynomials + null battery (Rem. 4.13)
<code>eisenstein_f21_residue_field_isomorphism.py</code>	Residue-field model of F_{21} , 21 checks + controls (Thm 4.19)
<code>eisenstein_norm_gte_constants.py</code>	Pre-registered Φ_6 /Eisenstein-norm scan of 20 GTE constants (§4.10.3)
<code>eisenstein_variety_point_count_prime_powers.py</code>	$\Phi_6(q)$ counts on 23 prime powers; torus orbits (Thms 4.20, 4.21)
<code>phi6_ladder_identities_and_nulls.py</code>	Identity web I1–I4 + rival-form and wrong-target nulls (Rem. 4.23)
<code>artin_mazur_zeta_gte_poly.py</code>	Periodic-point counts and cycle spectra, $n = 3$ – 7 (§6.7)
<code>gte_poly_glider_cycle_structure.py</code>	σ -equivariant glider decomposition of the $n = 7$ cycles (§6.7)
<code>gte_zeta_moebius_fixed_point_theorem.py</code>	Möbius orbits, de Bruijn certificate, general- q scan $q < 200$ (Thm 6.1)
<code>gte_zeta_drift_scaling.py</code>	Drift-3/7 mechanism tests; ether tile occupancy (Rem. 6.2)
<code>gte_zeta_period475_linear_structure.py</code>	$\sigma = T^{190}$, $T^{95} = \sigma^3$, spectral support census (Thm 6.3)
<code>gte_zeta_torus_entropy.py</code>	Spacetime-torus transfer operators; zero-entropy result (§6.7)
<code>cycle_spectrum_null_battery.py</code>	Null battery: 30 random + 10 structured GF(7) rules (§6.7)
<code>nineteen_factor_crt_tower_structure.py</code>	CRT split, induced 25-ring CA, twist orders, gauge periods (Thm 6.3)
<code>nineteen_factor_linearization_no_go.py</code>	$\text{charpoly}(DF) = x^3(x-2)^2$ at all 475 points (Thm 6.5)
<code>nineteen_factor_generalization_battery.py</code>	Pre-registered cross-field value-law battery (0/4) (§6.8)
<code>nineteen_factor_null_d_census_extension.py</code>	Extended 24-rule clock census ($d = 19$ at generic rate) (§6.8)
<code>triangle_lift_theorem.py</code>	Orbit-parity provenance; dual-method 7^8 lift census; flattening floor; field-generality sweep (Thm 4.4)
<code>triangle_residual_tests.py</code>	Identity-orbit multilinear inconsistency; projection nulls; 120-ordering census; coefficient-grammar null (§4.4, Thm 4.8)
<code>triangle_projection_battery_regenerate.py</code>	Four-projection battery artifact, pinned monomial ordering (Rem. 4.7)
<code>parity_projection_homomorphism_battery.py</code>	All 777 additive reductions: statuses, forced survivors, neighbor and data-perturbation nulls (Rem. 4.7)
<code>parity_projection_local_function_battery.py</code>	Exhaustive + sampled local-function recoding classes; per-class parity controls (Rem. 4.7)
<code>parity_projection_architecture_filters.py</code>	Displaced-vacuum and closure filters over all recorded battery exceptions (Rem. 4.7)
<code>parity_projection_unrestricted_vacuity.py</code>	Explicit lookup-forcing certificates for 76 of 128 vacuum-transparent elementary rules (Rem. 4.7)

Note on Taichi. Do not include `from __future__ import annotations` in any file importing Taichi: it activates PEP 563 lazy annotation evaluation, which breaks Taichi’s kernel argument type inspection at runtime.

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